Natural Knowledge

(main edition)

spirals, memes, and art
Book 1 of *Natural Knowledge - a collection*

Freddi Van Gogh

Natural Knowledge (main edition): spirals, memes, and art

Book 1 of *Natural Knowledge - a collection*

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The companion reference, *Modern Alchemy*, book 3 of this collection, holds this book's concepts as a numbered glossary with a notation for combining them.

How this book was made

The ideas, the questions and the decisions in this book are the author's. The author wrote the notes, the riffs and the rough drafts, and set out a recipe for each part. AI assistants turned that material into prose, checked it against sources and against the series' fact checker, and raised questions where the notes were wrong or unclear. The author answered them, made the calls, and edited the result.

In the terms publishers now use, the text is AI-generated: an AI drafted it, working from the author's own notes, research and writing as its source material. The cover was also made with an AI assistant, drawn in code.

Chapter 1 sets out what this book is attempting and — more usefully — what it is not. Chapter 17 collects the limitations, the strongest objection to the whole account, and what would count against it. A sceptical reader may prefer to take chapter 17 early.

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Preface: How the vocabulary emerged

This work began with an attempt to understand how seemingly unrelated things might be connected.

My academic background includes a degree in biochemistry, a minor in computer science, and doctoral training in computational biology. Those disciplines gave me different ways to look at the world: chemistry as a system of interactions, biology as an evolving organization of matter, and computation as a way to understand information, patterns, and complex systems.

But the immediate inspiration for this book came from a much more personal experience.

During a difficult period associated with an adverse reaction to an allergy medication, I experienced what I can only describe as an unusually intense period of pattern recognition. Connections seemed to appear everywhere: between scientific ideas, religious narratives, mythology, language, evolution, storytelling, and the organization of life. I began writing constantly. Over time, I accumulated hundreds of pages of notes, observations, diagrams, and associations.

The experience had a spiritual quality. I sometimes felt as though there were a guiding hand behind the connections I was noticing. I also experienced vivid imagery and ideas while trying to fall asleep. My thinking became increasingly expansive and difficult to organize. I was making connections faster than I could evaluate them.

I have used the phrase **"messiah effect"** to describe one aspect of this experience: the subjective feeling of suddenly seeing how many things fit together, as if previously separate pieces of knowledge have become part of a larger picture. I do not mean that this is a recognized scientific or psychological phenomenon, that it confers special authority, or that the connections perceived during such an experience must be true. It is my name for a particular experience of connectedness and intense pattern recognition.

After the episode passed and I was no longer taking the medication, I returned to my notes. What had initially seemed like an enormous and increasingly disorganized collection of ideas contained recurring themes. Beneath the apparent chaos were repeated questions about persistence, selection, organization, information, feedback, specialization, and emergence.

I began to see that many of the same organizational patterns appeared in very different settings. Chemical reactions, biological systems, neural networks, language, economies, and cultures are not the same kinds

of systems, and they do not operate through identical mechanisms. Yet they can sometimes be understood using a shared vocabulary.

This raised a broader question: might human beings have an innate capacity for unusually expansive pattern recognition? Our ability to recognize similarities, transfer ideas between domains, imitate nature, and construct new combinations has demonstrably contributed to technological and cultural development. Some speculative ideas, such as the "stoned ape" hypothesis, attempt to explain aspects of human cognitive development through altered states, but they remain speculative rather than established explanations. Leonardo da Vinci provides a more concrete historical example of someone who drew connections across anatomy, engineering, art, mechanics, and the natural world.

The ability to see patterns is powerful, but it is not automatically reliable. Humans can discover meaningful relationships, but we can also perceive connections where none exist. A compelling pattern is the beginning of an inquiry, not proof of an explanation. This book therefore tries to distinguish established science from analogy, hypothesis, speculation, and personal interpretation.

I remain an empiricist, even though the experience that inspired this work felt spiritual. I do not want to dismiss the subjective meaning of that experience, but neither do I want to present it as evidence for a super-

natural mechanism. The feeling of a guiding hand remains a personal and philosophical question.

One of the ideas that emerged from this process was that many phenomena once explained through myths can now be understood through physical and chemical processes. Thermodynamics, chemistry, biology, and evolution provide explanations for many forms of organization and complexity. But explaining some phenomena does not mean that every question has been answered. There is still much we do not understand about matter, life, consciousness, and the universe.

The purpose of this work is not to present a new theory or a manifesto. It is to develop a vocabulary: a collection of concepts that makes it easier to describe recurring patterns across different levels of organization. It is an attempt to understand where we came from, how complexity accumulates, and why certain forms of organization persist.

It began as an experience of seeing connections. It became an effort to find the vocabulary to explain them — and, just as importantly, to distinguish the connections that withstand scrutiny from those that do not.

1
Purpose and central idea

This book explores recurring patterns of organization across matter, chemistry, biology, cognition, language, economics, and culture.

The central proposition is that complex systems often develop through combinations of variation, interaction, selection, retention, amplification, specialization, and transmission — and that these processes can produce increasingly organized structures and behaviours without anything directing them.

The objective is to build a vocabulary that helps describe:

- How matter organizes into increasingly complex structures.
- How chemical systems persist and become self-reinforcing.
- How life may have emerged from non-living chemistry.
- How biological systems develop compartments and distribute functions.

- How cognition represents and compresses information.
- How language and culture accumulate and transmit information.
- How social, economic, and religious systems become organized and persist.

A sequence recurs throughout:

Variation → search → selection → retention → amplification → specialization → compression → transmission → emergence

This is a conceptual sequence rather than a law. Its stages overlap, repeat, run in parallel, and are sometimes absent entirely.

What this is not

Books of this shape usually leave their limits to the end, if they state them at all. Putting them here instead is deliberate, because they change how the rest should be read.

This is not a theory. A theory makes predictions that distinguish it from alternatives, and this framework does not yet do that. Chapter 17 states the objection in full and concedes most of it. What this book offers is a vocabulary — a set of terms that make certain transitions easier to describe and compare. That is less than a theory and not nothing.

It does not claim one mechanism explains everything. The same word can be correctly applied to a chemical reaction network and to a language without the two sharing any machinery. Selection in a population of molecules, in a population of organisms, and in a population of ideas are three unrelated processes that a single word happens to fit. The book returns to this warning more often than to any other point, because the warning is the thing most likely to be forgotten.

It does not fill the gaps it finds. The largest one is named in chapter 8: how chemistry first came to copy a pattern reliably enough for selection to accumulate across generations is not known, and nothing here narrows it. Chapter 17 lists the others.

It does not settle what it is not equipped to settle. Explaining how an idea spreads says nothing about whether the idea is true. This applies throughout, and it applies to religious and philosophical questions in particular, which chapter 15 examines as forms of cultural organization while declining to assess them as claims.

How claims are marked

A framework spanning this many fields will state things that are textbook settled, things that are actively disputed, and things that are the author's own reading. Presenting all three in the same confident voice would be the easiest way to mislead, and the hardest to detect.

So they are separated, and the reader should be able to tell which is which at every point:

Established. Stated plainly, with no hedge. The structure of the stratum corneum, the redundancy of the genetic code, the fact that mitochondria descend from bacteria.

Contested. Named as disputed, with the dispute described rather than suppressed. The RNA world. The cause of the genetic code's error tolerance. Whether the effects of vocabulary on perception generalize.

The author's own. Marked explicitly, usually in italics and usually set apart. The reading of early Earth as a vast parallel trial. The suggestion that life may have begun in more than one place. The entire cosmological aside in chapter 11.

Where a well-known claim turns out not to survive scrutiny, the book says so rather than quietly substituting a better one — the Inuit snow vocabulary in chapter 14 and the appendix in chapter 9 are both handled that way. Those passages are not digressions. They are the method being shown rather than asserted.

A borrowed form

The shape of this type of text structuring came from an unlikely place, and saying so seems better than letting it look like an accident. The Talmud.

The Talmud is built out of disagreement and does not tidy it away. Its core layer records legal rulings along-

side the positions that lost, attributed by name. Later layers argue with that layer and with each other, and a printed page sets commentators separated by centuries around the same text, visibly disputing it. Rejected views are kept rather than deleted — the tradition gives reasons for this, among them that a later generation may find the minority position was right, and that a ruling arrived at over an alternative should show the alternative it was chosen over. Some questions are marked as unresolved and left that way. Disagreement pursued seriously is treated as a thing of value in itself rather than a failure to reach agreement.

That is a form, and it turned out to be the right one for this material.

It is why chapter 8 gives three competing accounts of why the genetic code is error-tolerant and declines to pick, rather than presenting the most popular one as settled. It is why chapter 14 states the failed replication of the study its own argument was built on, in the body text, instead of quietly swapping in better evidence. It is why chapter 17 quotes the strongest objection to this entire framework at full strength before answering it, and concedes most of it. And it is why several questions here are left open in the text rather than resolved for the sake of a clean ending.

The limits of the borrowing should be stated plainly, because the two enterprises settle questions by completely different means. The Talmud reasons from

received texts and the authority of transmission; this book is answerable to evidence, and when evidence arrives, any previous position does not get to keep its seat out of respect. A failed replication is not a minority opinion awaiting rehabilitation. The resemblance is in the bookkeeping, not in the epistemology.

What was borrowed is narrower than it sounds and more useful than it sounds: the habit of recording who thought what and why, keeping the losing argument legible, and letting a question stay open on the page when it is open in fact. A great deal of scientific writing does the opposite — presents a consensus with its disputes sanded off — and something is lost when it does, because a reader given only the conclusion has no way to tell a settled question from a contested one that was summarized confidently.

The bookkeeping is not peculiar to religious law. In many courts a judge who disagrees with a ruling can publish a dissenting opinion beside it, and a later court sometimes adopts reasoning that first appeared in dissent. Courts that follow precedent explain each decision against the earlier cases it follows or departs from, so the reasons travel with the ruling. Many legislatures keep a verbatim record of their debates, with the amendments that failed and how members voted, so the case against a law stays on the record after the law passes. These institutions settle questions by authority and by vote rather than by experiment, so the limits

stated above apply to them too. What they share with the Talmud, and with this book, is the record.

How the book checks itself

Making claims, and marking them, is a promise, and promises drift. A manuscript of this length, revised over many passes, can state the same quantity two ways in different chapters, let a conclusion solidify as it is summarized, or lean on a premise it elsewhere calls contested. To catch that, the book is accompanied by a fact checker written in **Prolog**, a logic programming language.

A Prolog program is not a sequence of instructions. It is a set of **facts** and **rules** — Horn clauses, in the logician's terms — against which the system answers **queries** by searching for a proof. A fact states that something holds: that chapter 9 gives the giraffe's laryngeal detour as about four and a half metres. A rule states what follows from what: that two passages assigning incompatible values to the same quantity constitute a conflict, or that a claim presented as settled which depends, directly or through a chain of other claims, on one presented as contested is overstating its support. A query asks whether any conflict exists. The system answers by **unification**, matching the query's variables against facts and the heads of rules, and by **backtracking** through every way the rules could be satisfied, collecting each solution rather than stopping at the first.

Two properties suit this to the job. The search is exhaustive: every fact is tried against every rule, including the tedious combinations a human reader skims. And it operates under a **closed-world assumption** — whatever is neither stated nor derivable is treated as false — so a step the book never supports registers as missing, instead of being filled in charitably the way a sympathetic reader fills it in.

The checker reads three layers. Every sentence of both editions — this one and a shorter high-school edition — is captured automatically as a fact, with its numbers, dates, named authorities, and tier marking. The concepts and worked chains of *Modern Alchemy* are captured as relations — "can produce," "is a kind of," "is analogous to" — so that the companion's one-line summary of the book's arc can be tested link by link against the chains that are supposed to support it. And the central propositions are encoded by hand as structured claims, with their values, sources, dependencies, and arithmetic, each anchored to a verbatim quotation so that no claim can silently outlive the sentence it describes.

On its first run the checker found, unprompted, the gap that chapter 8 acknowledges: no worked chain carries a protocell to heredity. It also flagged a place where the book's own summary stated a conclusion more firmly than the chapter it rests on; the summary has since been corrected. What it cannot do is establish that

a claim is true. It tests whether the book's statements are consistent with one another and with the sources the book cites, and an error that is consistent everywhere passes. That is the limit this book describes for frameworks in general, applied to its own machinery.

The rules, the encoded claims, and instructions for running the checker are collected in *Prolog Logicbase*, book 4 of this collection.

How to read it

The chapters run roughly from physics to culture. That ordering is a convenience for exposition and not a claim that the later levels reduce to the earlier ones, or that history had to run this way.

It is also not strictly chronological, and one departure is worth flagging. Chapters 5, 6, and 7 follow boundaries as a theme — compartments, then compartments inside compartments, then an organism and the organisms living on it — which means chapters 6 and 7 describe things that already require heredity, while chapter 8, where heredity arrives, comes after them. The thematic grouping seemed worth the cost. Chapter 16 sets the sequence out in its actual order.

Modern Alchemy, the companion reference, holds these concepts as a numbered glossary with a notation for combining them. It is more useful alongside the book than before it.

A note on what the book is for. It is not an argument that is supposed to end in agreement. It is an attempt to make a set of transitions describable — from chemistry that persists, to chemistry that copies, to systems that model, to systems that tell each other what they have modelled — using terms that travel between them without pretending the territory is the same. Where the terms travel well, something has been gained. Where they travel badly, that is worth knowing too, and the book tries to be useful about both.

2
A recurring vocabulary of emergence

A small number of concepts recur across every domain this book covers. They are set out here so that later chapters can use them without redefining them, and so that the reader can hold the whole set in view at once.

A warning belongs before the list rather than after it. These terms travel well, and that is exactly what makes them dangerous. A word that correctly applies to a chemical reaction and to a language has not shown that the two have anything in common beyond being describable. A later section of this chapter takes that seriously enough to work through an example.

The terms

Variation. The production of differences within a system. Variation arises through random events, environmental fluctuation, recombination, mutation, experimentation, or deliberate action.

Search. The exploration of possible states or combinations. Search requires no intention, no intelligence, and

no objective. A mixture of molecules bumping into one another is searching, in the only sense this book needs.

Selection. The differential persistence or success of some configurations over others. Selection can be imposed by environmental conditions, physical stability, resource availability, competition, compatibility, or any other constraint that treats configurations unequally.

Retention. The persistence of a configuration over time — through chemical stability, feedback, replication, memory, physical structure, or institutional practice.

Amplification. A process by which an existing pattern or difference becomes more abundant or more influential. Positive feedback amplifies small initial differences, sometimes enormously.

Concentration. The local accumulation of matter, energy, or information. Processes that are negligible at low density can dominate at high density, so concentration frequently decides what is possible.

Compartmentalization. The formation of a boundary that localizes processes and permits conditions inside to differ from conditions outside. Chapter 5 is about why this changes so much.

Metacompartmentalization. Compartments nested within, or integrated into, larger compartments. Chapter 6.

Coupling. The linking of processes so that one supports, regulates, or enables another. Coupling is how separate things become one system.

Specialization. The development of components that perform particular functions. It improves what the whole can do and costs each part its independence.

Compression. The reduction of complex information into more manageable representations. Compression makes information easier to store, process, and transmit, and necessarily discards something.

Transmission. The movement of information, structure, or behaviour between components, organisms, generations, or populations.

Accumulation. The retention and layering of change over time. Accumulation is what makes a history rather than a sequence of unrelated states.

Feedback. A process in which a system's output influences its subsequent behaviour. Feedback stabilizes, amplifies, regulates, or destabilizes, depending on its sign and strength.

Emergence. The appearance of system-level properties arising from interactions among components — properties not apparent from examining any component alone.

Selection bias. Observing what persisted rather than everything that was tried. This belongs in the list because it is a property of the observer rather than of the system, and because it distorts every question in this book. Chapter 3 shows how severely.

What the list is for

These are descriptive terms, not explanatory ones. Saying that a system underwent variation and selection says what happened at a level of abstraction that deliberately omits the machinery. That abstraction is useful for noticing that two situations rhyme. It is useless for establishing that they work alike, and the step from one to the other is where arguments of this kind usually fail.

The terms are also not independent. Compression tends to follow specialization, because a specialized component does not need the information a general one does. Amplification requires feedback. Accumulation requires retention. The vocabulary describes a connected set of processes rather than a checklist.

The same word, three mechanisms

The clearest way to hold the warning is to take one term and watch it change underneath.

	What varies	What selects	How the result persists
Chemistry	Molecular configurations produced by collision and reaction	Thermodynamic stability and reaction kinetics — which structures survive the next perturbation	Nothing is copied. The structure either lasts or it does not.
Biology	Genetic sequence, through mutation and recombination	Differential survival and reproduction in a population	A template is copied with high fidelity into offspring.
Culture	Ideas and practices, through misremembering, reinterpretation, and deliberate change	Whether something is attended to, understood, remembered, and repeated	Reconstructed by an observer who may have understood it incompletely.

Three rows under the heading "selection." The processes share no machinery. Chemical persistence involves no population and no copying. Biological selection requires a replicating template and operates across generations. Cultural selection operates within a single lifetime, passes between unrelated people, runs in any direction including backwards in age, and copies with such low fidelity that the thing transmitted is frequently not the thing that was sent.

Any sentence applying "selection" across these rows is doing less work than it appears to. What the shared term buys is the question — *is something being varied, and is something treating the variants unequally?* — which is worth asking in all three cases. The answer is different every time, and the answer is the part that matters.

The companion

These concepts and roughly a hundred more are set out in *Modern Alchemy*, the companion volume, with permanent identifiers, one symbol each, and a notation for combining them. It is deliberately oversimplified and says so on its first page.

The notation exists to make combining easy. A field cannot be held in the head, but a handle can, and handles can be set side by side. Each concept gets a symbol, and a few operators join them: + for things acting together, → for "can produce, under conditions not stated," ≈ for a resemblance that claims no shared mechanism. An argument that takes pages in chapter 5 then fits on a line — `Ap + Df → V`, amphiphiles diffusing in water can close into a vesicle — and that line can be laid beside a line from another field to see whether they rhyme. Most cross-domain questions begin as exactly that kind of naive juxtaposition. The book's whole arc, from thermodynamics to the evolution of religion, fits in a few lines of it.

The name is a joke with a point in it. Alchemy had a working notation, real apparatus, and a conviction that the right combinations would reveal how matter worked. It was wrong about nearly everything, and chemistry grew out of it anyway. Its symbols also combined as smoothly for a transmutation that never worked as for a distillation that did, and the same holds here: symbols always compose, and the world is under no obligation to follow. So a formula is a question rather than a finding. Analogies are written with ≈ and kept away from the arrow on purpose, because an analogy taken for a mechanism is the error this book most needs to avoid. And a formula is silent on rates, probabilities, scale, and reversibility, and records only what persisted.

It is most useful alongside this book rather than before it, and its closing sections, on what the notation cannot express and on using it without being fooled, make the same point this chapter has just made, at greater length and with less patience. The fact checker described in chapter 1 reads its worked chains as well, and tests that every step of the arc either has a chain behind it or is named as a gap.

3
The early Earth as a chemical search space

Before life existed, Earth contained an enormous variety of chemical environments.

Sunlight, heat, lightning, volcanic activity, minerals, water, tides, and atmospheric chemistry created many opportunities for molecules to interact. Day–night cycles, seasonal changes, wet–dry cycles, temperature gradients, and differences in chemical composition continually altered local conditions.

Molecules moved through diffusion and **Brownian motion** — the ceaseless jostling that thermal energy imparts to anything small enough to be pushed around by it. Some accumulated in particular environments; others dispersed. Surfaces, pores, mineral structures, and interfaces could concentrate molecules or influence which reactions occurred.

This created an enormous parallel search space.

Three engines

None of that chemistry would have gone anywhere without a sustained supply of energy. Early Earth had three largely independent sources. Each pushed in one direction, and none of them was going to run out.

The Sun delivered ultraviolet radiation and heat. Ultraviolet light carries enough energy to break and form chemical bonds directly, and the day–night cycle meant it arrived in pulses rather than at a constant rate.

The planet's interior supplied heat, chemically reduced minerals, and hydrothermal circulation. Volcanic systems produced steep thermal and chemical gradients where hot volcanic lava met cold water, and lightning in ash plumes added bursts of electrical energy of the kind later shown to drive organic synthesis in the laboratory.

Volcanoes also supplied raw material: sulfur dioxide and hydrogen sulfide, which dissolve in water as sulfite and sulfide, and silica-rich hot-spring water that crusts surfaces with sinter. The same lightning, like that of ordinary storms, turned nitrogen from the air into nitrogen oxides, which end up in water as nitrite and, mostly, nitrate. Coarse ash blankets the ground near the vent; sulfur gas lofted high becomes sulfate droplets that stay aloft for about a year, spreading on the wind. Whether these compounds were at concentrations to matter is still contested, although one can imagine smaller pools

and lakes accumulating meaningful levels more so than the ocean.

The Moon supplied motion on a schedule. Tides stirred shallow water, flooded margins and drained them again, and did so predictably. The Moon orbited closer to Earth's surface then and Earth's day was shorter, so tides arrived more often and reached further than they do now.

The three together matter more than any one of them. A system can build and hold local organization while energy flows through it and waste heat leaves — this is what a **dissipative structure** is, and it is ordinary physics rather than an exception to it. What early Earth had was not a single energy source but a continuous, varied, and rhythmic supply, which is exactly the condition under which such structures form and persist.

The arithmetic of deep time

"Billions of years" is the phrase that usually carries the weight in explanations of life's origin, and it carries it poorly. The number is too large to picture, so it functions less as an argument than as a request to stop asking. It also points at the wrong quantity.

The useful question is not how long Earth has existed. It is **how often the experiment ran**.

Consider a shallow pool at a tidal margin. It is flooded, agitated, partly dried, and refilled. Amphiphilic molecules in it — the kind with a water-attracting end

and a water-repelling end — assemble into films and droplets, hold together for a time, and are broken apart again. Each assembly is a trial: a slightly different mixture, enclosed for a slightly different interval, under slightly different conditions.

Suppose one such cycle completes every six minutes. Then that single pool runs:

	trials
per day	~240
per year	~88,000
per century	~8.8 million
per ten million years	~10^{12}

Ten million years is a rounding error in Earth's history. One pool.

Earth is about 4.54 billion years old, so its history holds about 450 ten-million-year spans.

The six-minute figure is invented, and the honest thing is to say so and then show that it does not matter. Make the cycle an hour instead and the trial count falls by a factor of ten, to around 9×10^{10}. Make it a full day and it falls by a factor of 240, to around 4×10^{9}. Even at the most pessimistic end that is billions of trials, in one pool, over an interval geology would not bother to name. Across any plausible cycle time the conclusion is the same, which is a sign the argument rests on the structure of the situation rather than on a number someone chose.

And there was a planet's worth of such places — tidal margins, hydrothermal vents, hot springs, pores in wet rock, films on mineral surfaces — all running at once, independently, for hundreds of millions of years.

So the picture is not one slow process grinding away for four billion years. It is an extremely large number of short, parallel, independent, cheap experiments, nearly all of which failed within minutes, running simultaneously across an entire planet. Deep time supplies the opportunity. Cycling supplies the throughput. The second is doing more of the work than it usually gets credit for.

This way of framing the problem is the author's, and the figures above are an order-of-magnitude illustration rather than a measurement. The underlying components — wet–dry cycling, spontaneous assembly of amphiphiles, the role of mineral surfaces — are well established and are taken up again in chapters 4, 5, and 17.

Constrained, not random

A familiar image for this kind of process is the one about enough monkeys at enough typewriters eventually producing Shakespeare. It is useful for conveying how large numbers make improbable outcomes ordinary, and it is misleading in one important respect.

Chemistry is not a random typist. Molecular properties, bond energies, reaction rates, concentrations, temperature, and the geometry of available surfaces all

strongly constrain what can happen. Most of the possible arrangements of a given set of atoms are never explored at all, because they are not reachable from where the system actually is. The search is vast, but it is not uniform, and it is not blind in the way the image suggests.

This matters because it cuts the problem down. A process that samples randomly from everything possible would need far more time than Earth has had. A process whose next steps are heavily constrained by the chemistry already present is exploring a much smaller space — and exploring it in parallel, in enormous numbers, continuously.

Nothing in this requires foresight. Physical conditions created possibilities, and the properties of the resulting chemical systems determined which ones lasted long enough to matter.

What cannot be seen from here

One caution belongs at the end of any argument of this shape.

What is visible from the present is the chemistry that led to life. The pathways that failed, stalled, or were destroyed left nothing behind to examine. This is **selection bias**, and it is unusually severe in this case, because the surviving lineage is the only one that produced observers capable of asking about it.

That a pathway persisted does not establish that it was likely, and it certainly does not establish that it was inevitable. A planet running an enormous number of parallel experiments will produce some improbable outcome; the one it produced is the one being looked at. How probable life was, given early Earth's conditions, remains genuinely unknown. The sample size is one.

4
From chemical reactions to self-sustaining networks

Chapter 3 described an enormous number of chemical trials running in parallel. Most produced nothing that lasted. This chapter is about the ones that did, and about what "lasting" can mean for a reaction.

A chemical reaction can produce a compound that participates in further reactions. Some reactions reinforce the production of their own components, or help sustain a whole network of reactions. Several ideas are needed to describe how:

- **Catalysis:** a substance increases the rate of a reaction without being consumed in the overall reaction.
- **Autocatalysis:** a reaction product helps catalyse the production of more of itself.
- **Autocatalytic sets:** networks in which the components collectively sustain the network, with no member required to catalyse itself.
- **Reaction networks:** connected sets of reactions in which products and reactants participate in multiple processes.

- **Dissipative structures:** organized patterns that arise and persist in systems driven away from equilibrium by flows of energy or matter.

Those are definitions. The rest of this chapter is given over to the demonstrations, because each of these is something that has been observed rather than something the argument needs to be true.

What catalysis buys

A catalyst does not make an impossible reaction possible. It makes a slow reaction fast, by offering a path with a lower barrier.

The distinction matters for the early Earth. Reactions that are thermodynamically allowed but take ten thousand years at ambient temperature might as well not happen in most places, because few conditions hold still that long. A mineral surface or a small molecule that drops the barrier turns a reaction that effectively never occurs into one that occurs constantly.

Catalysis therefore does not change what chemistry is possible. It changes which possibilities are reachable within the time a given environment stays put.

Environment is doing quiet work in that sentence. It names a concept people made, not a unit nature hands over. The skin of a face is an environment for the bacteria living on it. So is a yard, and so are a house, a town, a region, and the whole of nature, overlapping

and nested, with edges drawn wherever it is convenient to draw them. The timescale is as loose as the edges. How long an environment stays put depends on where its line is drawn, and the only measure that matters is whether there was enough time for something to happen. For a tidal pool, splashed by waves and refilled by tides, that may be minutes or hours; for water held in a pore deep in rock it may be far longer. A catalyst matters because it brings a reaction inside whichever window applies.

Autocatalysis, demonstrated

Autocatalysis is the point where a reaction starts to have something like an interest in its own continuation, and it is worth seeing concretely rather than taking on trust.

The **formose reaction** is the standard prebiotic case, known since Butlerov described it in 1861. Formaldehyde, a simple molecule plausibly abundant on the early Earth, reacts with itself in the presence of a base to produce sugars. The reaction is autocatalytic: one of its early products, glycolaldehyde, catalyses the process that produces more of it. The characteristic behaviour is a long sluggish period in which almost nothing happens, followed by a sudden acceleration as the product that speeds the reaction begins to accumulate. The reaction builds the thing that makes it faster.

That shape — slow, then abruptly fast — is the signature of autocatalysis, and it is the same shape as a population growing or a rumour spreading.

The second demonstration is more direct. In 1996 a research group reported a **peptide** (a short protein-like chain) 32 amino acids long — its structure borrowed from a small unit of an ordinary yeast protein — that catalyses its own formation. Presented with two shorter fragments corresponding to its own halves, it binds them against itself and accelerates their joining into another complete copy. A molecule making copies of itself, built from ordinary amino acids, with nothing biological involved. Comparable systems had already been built from short nucleic acid strands a decade earlier, and more capable ones have been built since — pairs of RNA enzymes that make each other, multiplying exponentially with no proteins involved.

These are laboratory systems, designed rather than found, and that limits what they establish. They do not show that such molecules arose on the early Earth.

The newest result narrows that limit without removing it. In 2026 a group reported an RNA enzyme only 45 nucleotides long that copies RNA templates. It began as a find rather than a design: it was selected from pools of random sequences, then improved by further rounds of laboratory evolution and engineering. Working in mildly alkaline ice, from building blocks three letters long, it can make both its complementary strand and a

copy of itself, which are the two halves of a replication cycle. It does them only in separate reactions, though, with yields of about 0.2 per cent after 72 days and about 6 per cent of letters copied wrongly, and the paper's own model finds that at this accuracy almost none of the copies would still work. Its discoverers note that 45 letters approaches the lengths unaided chemistry may be able to string together, and read the find as a sign that copying molecules are commoner among random RNA sequences than had been thought. They are equally clear that faster and more accurate copying is still needed before replication could sustain itself.

There is a second limit, more interesting than the first. The 1996 peptide's reproduction is **parabolic** rather than exponential: the rate rises with the square root of how much template is present, not in proportion to it. The reason is that copies tend to stick to one another, and a molecule bound to its own product is not available to catalyse anything. Parabolic growth has a consequence that matters for everything downstream — a faster variant cannot drive a slower one to extinction, because its advantage shrinks as it becomes common. Competition of the kind evolution requires does not get started. The same difficulty appears in most self-replicating chemical systems built since.

So what these experiments establish is narrower than it first appears, and still substantial. Self-reproduction is a chemical property available to fairly simple molecules,

and not something that had to wait for life to invent it. Self-reproduction *good enough to support open-ended selection* is a further step, and it is one nobody has yet demonstrated.

Sets that sustain themselves collectively

A molecule that catalyses its own production is one route. There is another, and it may be more plausible, because it does not require any single molecule to do anything remarkable.

Consider a set of molecules in which A catalyses the formation of B, B catalyses C, and C catalyses A. No member catalyses itself. The set as a whole catalyses itself. Given a supply of raw material, it persists and reproduces its own composition, and the property of self-maintenance belongs to the network rather than to any part of it.

Stuart Kauffman argued in 1986 that such **collectively autocatalytic sets** should appear almost unavoidably in sufficiently large and varied chemical mixtures — that past some density of molecules and reactions, the probability of a closed catalytic loop existing somewhere in the mixture approaches certainty. The argument has since been formalized and explored computationally.

The status of this is worth stating precisely, because it is a case where the mathematics is stronger than the chemistry. The theoretical result is solid: in a sufficiently

rich network, self-sustaining subsets are expected. Whether real prebiotic chemistry was rich enough, and whether the catalysts available were specific enough, is not settled. Experimental systems demonstrating collective autocatalysis exist but are small and mostly constructed rather than discovered.

This is also the first appearance of something the rest of the book returns to repeatedly: a property that belongs to a set of components and to none of them individually.

Order that lasts only while it is driven

The last idea is the one that explains why any of this persists.

Heat a shallow layer of liquid evenly from below. At small temperature differences, heat passes through by conduction and nothing visible happens. Past a threshold, the liquid abruptly organizes itself into a tiling of convection cells, often hexagonal, each circulating in a fixed direction. The pattern is orderly, it is not imposed from outside, and it appears spontaneously.

Stop the heating and it does not freeze in place. It disappears.

That is a **dissipative structure**: organization that exists because energy is flowing through, and only for as long as it is. Ilya Prigogine's work on these systems, recognized with a Nobel Prize in 1977, is the reason the phrase exists. The crucial point is that the structure is

not a thing that was built and now persists. It is a process that maintains an appearance of being a thing.

Chemistry has its own version. The Belousov–Zhabotinsky reaction, in a shallow dish, cycles between states and generates travelling waves and rotating spiral patterns that can continue for an hour or more before the reagents are exhausted. Its mechanism includes an autocatalytic step. It was initially disbelieved — a chemical reaction that oscillates rather than settling looked to reviewers like an error — and it is now a standard demonstration. (The spirals it produces are the fifth category in chapter 10 — a wave front curling around its own free end, with nothing material rotating at all.)

The same description applies to every organism, which is the reason the concept appears here rather than in a chapter on physics. A cell is not a structure that was assembled and now endures. It is a set of processes that rebuild it continuously, and it persists for exactly as long as they continue.

A possible progression

Putting these together, a possible sequence:

1. Chemical exploration produces many molecular combinations.
2. Some reactions generate compounds that participate in further reactions.

3. Catalysis makes particular pathways fast enough to matter on environmental timescales.
4. Interconnected reactions begin to reinforce one another.
5. Some networks persist longer than isolated reactions.
6. More stable networks become coupled to energy capture, containment, and eventually information retention.

The steps are not equally supported. Steps 1 through 3 are directly demonstrated: mixtures that diversify, products that feed further reactions, catalysts that make particular pathways fast. Steps 4 and 5 have been shown only in the laboratory, in systems that are small and mostly designed rather than found, and whether early Earth's chemistry was rich enough to produce them unaided is, as above, not settled. So the sequence as a whole is a conjecture, and stays a conjecture until the chemistry is better understood. Step 6 is where the account becomes genuinely incomplete.

What this does not solve

Three honest limits, since a chapter that presents only the encouraging results is not worth much.

The formose reaction, the usual example, has a serious problem as an origin-of-life mechanism: it is indiscriminate. It produces a complex mixture of sugars and

degradation products — described in the literature, not unfairly, as tar — in which the biologically relevant sugar is a minor component and is unstable besides. Selectivity, not reactivity, is the difficulty. Some minerals and borate compounds improve matters, and whether they improve them enough is unresolved.

The self-replicating molecules described above were designed, or, in the newest case, selected from random sequences and then refined by chemists. Showing that a chemist can build or find one does not show that chemistry can find one unaided, and the search space from chapter 3, large as it is, is not self-evidently large enough.

Most importantly — and this is the parabolic-growth problem generalized — **no demonstrated autocatalytic system evolves open-endedly.** They sustain themselves and they copy themselves. What has not been shown is a chemical system in which better variants keep displacing worse ones, so that improvements accumulate across generations without limit. That is what separates persistent chemistry from life, it is the subject of chapter 8, and it remains the largest unfilled step in this account.

5
Compartments: creating local systems

One important transition in the development of life may have been the emergence of compartments.

A compartment separates a local chemical environment from its surroundings. This can happen through physical boundaries such as lipid membranes, mineral pores, or other structures.

The central fact about such boundaries is that the simplest of them require nothing to build them.

What self-assembly looks like

Oil and vinegar shaken together in a bottle disperse into droplets and then separate again within minutes. Oil and water do not mix, and shaking only postpones the outcome.

Add egg yolk and the result is mayonnaise, which does not separate.

The difference is a class of molecule present in the yolk. **Amphiphiles** have two ends with opposite preferences: one that associates readily with water, one that

does not. A molecule like that cannot be comfortable anywhere in a mixture of oil and water except at the boundary between them, with each end in the phase it prefers. Enough of them arrive at that boundary, coat the oil droplets, and the droplets stop merging. Nothing assembled them. Each molecule found its lowest-energy position, and a stable structure is what the sum of those positions looks like.

Soap works by the same principle in reverse. Above a certain concentration in water, soap molecules cluster with their water-repelling ends turned inward and their water-attracting ends facing out. The result is a **micelle** — a ball of tails wrapped in heads. Grease dissolves into the interior, which is why the technique works.

A micelle is not yet a compartment in the sense that matters here, and the distinction is easy to lose. A micelle has no interior space: its centre is a knot of hydrocarbon tails with nowhere for water to be. Nothing dissolved in water can be contained inside one.

The structure that does contain something is a **bilayer vesicle** — two sheets of amphiphiles back to back, tails meeting in the middle, water-facing heads on both outer surfaces, the whole thing curved around and closed. That arrangement encloses a volume of water, separated from the water outside by a film two molecules thick.

Which structure forms is largely a matter of molecular shape. Amphiphiles with one tail tend to pack into

micelles. Those with two tails, which cannot taper to a point, pack into sheets instead, and sheets close into vesicles. The phospholipids in every cell membrane have two tails.

This matters for the origin of life because the simpler molecules are the plausible ones. Fatty acids are short, single-tailed, and form readily under prebiotic conditions and in meteorites. Under the right concentration and acidity they also form vesicles rather than micelles. Laboratory work has shown such vesicles growing by taking up more fatty acid, dividing when pushed through narrow pores or agitated, and enclosing larger molecules including RNA while they do it.

Why a boundary changes what is possible

Compartments provide several advantages:

- They concentrate molecules in a limited space.
- They reduce the dispersal of useful reaction products.
- They create differences between internal and external environments.
- They allow chemical networks to interact repeatedly.
- They make it possible for local conditions to be maintained.
- They create units that may persist, grow, divide, or be selected as a whole.

The last of these does more work than the others. The first five improve the chemistry. The sixth changes what the chemistry is subject to.

Selection before anything is alive

Without a boundary, a productive reaction is a local event whose products diffuse away and benefit nothing in particular. With a boundary, the products stay with the system that made them, and systems can be compared.

That comparison does not require heredity, genes, or life. It requires only variation among compartments and differences in how long they last — which is the minimum condition for selection stated in chapter 2.

There is a demonstrated version of this that is worth describing, because it shows the mechanism rather than asserting it. Fatty acid vesicles are in constant exchange with their surroundings and with each other; molecules leave the membrane and join other membranes. A vesicle containing more dissolved material has higher osmotic pressure inside, which places its membrane under tension. A membrane under tension takes up fatty acid more readily. The consequence, observed in the laboratory, is that vesicles with more encapsulated content grow — and they grow at the expense of emptier vesicles nearby, which shrink as their membrane molecules migrate.

A compartment that happens to contain more is therefore better at acquiring the material it is made of. No information is copied, nothing is inherited, and no organism is involved. Physical competition between compartments is already underway, driven by osmotic pressure and membrane mechanics alone.

This is as close as chemistry comes to selection without life, and it arrives early, from the geometry of amphiphilic molecules and the behaviour of water.

Boundaries that store energy

A membrane also does something beyond containment. It can hold a difference in the concentration and electrical charge of ions between its two sides — an **electrochemical gradient**.

A watermill is a reasonable way to picture this, kept as a picture rather than a mechanism. A watermill does no work on its own; it extracts work from water moving down a height difference that something else maintains. A membrane holding more of a given ion on one side than the other maintains a difference of the same general kind, and processes in the membrane can be driven by letting that difference discharge. Instead of water and differences in elevation along a river, the energy is derived from charged ions and differential charges across a membrane.

This is central to how modern cells handle energy. Both respiration and photosynthesis work by pumping

protons across a membrane and then using the return flow to drive synthesis. Imagine the watermill again, with buckets of water carried by hand back to the top of the river so that more water flows through the mill and keeps it turning; the pumping is the carrying. The analogy is useful; the inference that early compartments possessed anything like this machinery is not available. Modern membrane bioenergetics is elaborate and evolved. What the earliest compartments plausibly had was the simpler precondition: a boundary across which a difference could exist at all.

Some proposals for the origin of life invert the usual order here and start from gradients that already existed — the natural differences in acidity and chemistry across the mineral walls of alkaline hydrothermal vents, where the geology supplies the gradient and the first membranes only have to take advantage of it.

Compartmentalization makes local chemical organization possible. It also makes local chemical organization comparable, which is the part that matters for everything after.

6
Metacompartmentalization, distributed labour, and emergence

Compartmentalization can happen more than once.

A compartment can contain smaller compartments, and multiple compartments can be organized into a larger one. This nested arrangement can be called **metacompartmentalization**, and it describes organization developing across several levels at once:

- Molecules form interacting chemical networks.
- Networks operate within compartments.
- Compartments develop into or contribute to cells.
- Cells organize into tissues.
- Tissues combine into organs.
- Organs operate together within organisms.
- Organisms interact within populations and ecosystems.

At each level the larger system can develop properties absent from any component alone.

Distributed labour and what it costs

Nesting creates the opportunity for **distributed labour**.

In a chemical network, different reactions contribute different functions. In a cell, different molecules and structures perform specialized tasks. In a multicellular organism, cells become specialized; in organisms with organs, tissues perform complementary functions.

Specialization increases what the whole can do. Rather than every component doing everything adequately, components become good at particular things.

It is also irreversible in a specific and consequential way. A component that becomes highly effective at one task generally loses the ability to function alone. A liver cell cannot live as a free organism. A mitochondrion cannot be cultured. The larger system must then coordinate its parts, move resources between them, regulate internal conditions, and manage conflicts between what suits a component and what suits the whole.

So the route to complexity runs:

Containment → interaction → specialization → interdependence → coordination → emergence

This is a pattern, not a progression. Greater complexity is not more stable or more successful; specialization introduces dependencies, and a failure in one component propagates. Many lineages have become simpler,

and parasites routinely discard entire organ systems their hosts can supply instead.

The major transitions

Evolutionary biology has a name for the events this chapter describes, and a short list of them.

John Maynard Smith and Eörs Szathmáry proposed in 1995 that the history of life is punctuated by a small number of **major transitions**, each sharing a common signature: entities capable of independent replication before the transition become capable of replicating only as part of a larger whole afterwards.

Their list has eight entries. It runs from replicating molecules becoming enclosed in compartments; through independent replicators becoming linked into chromosomes; through the division of labour between DNA as information and protein as function; through prokaryotic cells becoming eukaryotic; through asexual lineages becoming sexual populations; through single cells becoming multicellular organisms; through solitary individuals becoming colonies in which most members do not reproduce; to human societies organized by language.

The specific list is disputed, and the disputes are instructive. Critics have argued that the eight entries lack theoretical unity — that they were assembled because each is a large change rather than because they share a mechanism. The sexual-reproduction entry in

particular is questioned, since it is not obvious that it fits that signature. Some argue the appearance of nervous systems should be on the list and is not. The framework has been revised more than once since.

The signature is the part that survives the dispute, and it is the useful part. Each transition converts independent things into parts, and the conversion is what the word metacompartmentalization is pointing at. A free-living bacterium becomes an organelle. A free-living cell becomes a tissue cell. An organism that could reproduce becomes a sterile worker.

Something is given up every time, and it is always the same thing: the capacity to continue alone, exchanged for a dependency on cooperation.

Endosymbiosis

The sharpest documented case is the origin of mitochondria and chloroplasts.

The endosymbiotic account holds that both descend from free-living bacteria taken up by ancestral eukaryotic cells, which then developed permanent relationships with their hosts. Mitochondria trace to a group of bacteria related to modern alphaproteobacteria; chloroplasts to cyanobacteria.

The evidence is unusually direct. Both organelles retain their own circular DNA, distinct from the cell's nuclear genome. Both have their own ribosomes, and those ribosomes resemble bacterial ribosomes rather

than the host's — closely enough that some antibiotics targeting bacterial protein synthesis also affect them. Both are enclosed by a double membrane, consistent with having been engulfed. Both divide by splitting rather than being built from scratch.

The proposal was not well received. Lynn Margulis — publishing in 1967 under the name Lynn Sagan — recalled that the paper that set out the modern case was turned down by around fifteen journals before the *Journal of Theoretical Biology* took it. The idea remained marginal for years afterwards. Molecular sequencing eventually made it difficult to dispute. It is now standard biology, taught as settled.

The episode is worth remembering for the same reason the appendix is worth remembering in chapter 9: the confidence of a field at a given moment is not the same thing as the evidence. It is also worth not over-reading. A theory being rejected many times and later vindicated is a real pattern, and it is not evidence about any *particular* rejected theory. Most ideas that fifteen journals decline are declined correctly.

The most telling detail concerns where the genes went. The ancestral bacterium would have carried something on the order of a few thousand genes. The human mitochondrial genome carries thirty-seven. The rest were not discarded — the great majority migrated to the host cell's nucleus, where they remain, still producing

proteins that are manufactured in the main body of the cell and then imported back into the mitochondrion.

The organelle did not merely become dependent. It handed over the instructions for building itself. Integration at this level is not cooperation between two parties; it is the construction of a single system that can no longer be separated into the things it was made from.

Endosymbiosis also shows that new levels of organization need not arise by one lineage gradually specializing. They can arise by combination — two evolutionary histories merging into one.

What holds a level together

Every statement above has described nesting as though it happened and then stayed. It does not stay by itself.

A multicellular organism is a cooperative arrangement among cells, many of which remain, individually, capable of replicating. They carry machinery for division that predates animal multicellularity by a billion years. The organism works only because that capacity is held in check — by signals that tell a cell when to divide and when to stop, by programmed cell death removing cells that malfunction, by immune surveillance.

Cancer is what the arrangement looks like when the checks fail. A lineage of cells resumes doing what single cells do: proliferating as circumstances allow, without regard to the larger system. It is not an

invasion from outside. It is a reversion to the level below, by cells that never lost the ability.

The same tension appears at other levels. Stretches of DNA exist that are good at getting themselves copied and do nothing for the organism carrying them, sometimes at its expense. Mechanisms exist by which a gene can bias its own chances of ending up in a gamete, against the interests of the rest of the genome. In colonies of social insects, workers police one another's attempts to reproduce — an odd behaviour that leaves one queen bee laying nearly all of a hive's eggs while most of the population works and dies without ever reproducing.

The general shape is this: a higher level of organization persists only while it suppresses competition at the level below it, and the suppression is never complete. Every level of nesting is an ongoing arrangement rather than a settled fact, maintained against a lower level that has not forgotten how to act for itself.

This is the same structure as chapter 4's dissipative structures, one scale up. A convection cell persists only while energy flows through it. A multicellular organism persists only while it keeps putting down its own cells. Neither is a thing that was built and now endures. Both are processes that have to keep going to survive, almost like a tontine the cells never signed up for but are locked into all the same.

It also supplies the caution this chapter needs. Nesting is not an achievement that accumulates safely. It is a cost that has to be paid continuously, and when payment stops, the level below is still there — be it cancer, parasites, or death.

7
The human microbiome: an organism within a larger ecosystem

The human microbiome provides another important example of metacompartmentalization and distributed biological labour.

The human body is not a collection of human cells operating independently. It is also home to diverse microbial communities — bacteria, archaea, fungi, and viruses — occupying different environments including the skin, mouth, gut, and other sites.

Those populations are not distributed uniformly. Different parts of the body present different physical and chemical conditions, and different conditions support different communities.

The gut

The gut contains a complex microbial ecosystem whose members interact with one another, with food, and with the host. Microbial functions there include:

- Breaking down dietary compounds that human enzymes cannot fully digest.
- Producing metabolites usable by the host or by other microbes.
- Contributing to the development and regulation of immune responses.
- Competing with potential pathogens for space and resources.
- Transforming compounds, including some involved in drug metabolism.
- Participating in networks of metabolic exchange.

These functions are distributed across many organisms. One species may produce a compound another species consumes. The products of one metabolic pathway become the inputs of another. The community's overall activity therefore depends on interactions among organisms rather than on the capabilities of any single species.

This resembles the distributed labour of a multicellular organism, and the resemblance has a limit worth stating. Microbes are separate organisms with their own evolutionary interests. Their interactions include

cooperation, competition, and exploitation, and the balance shifts with conditions. Cells in a body share a genome and mostly share a fate. Members of a microbial community do not.

The skin: a barrier that evolved into an occupied world

The skin makes the same point from a different direction, and more sharply, because it is the surface where host and microbe actually negotiate.

Terrestrial life did not emerge into a sterile world. By the time the first vertebrates moved onto land, bacteria had been colonizing every available surface for billions of years — in biofilms on rock, in soil consortia, in mats along tidal margins. Skin as it now exists — layered, cornified, chemically selective — evolved *into* that environment. It did not evolve first and then get colonized.

That changes what the barrier is for. The outermost layer, the **stratum corneum**, is often described as bricks and mortar: flattened dead cells called corneocytes, embedded in a lipid matrix of ceramides, cholesterol, and free fatty acids. The usual reading is a wall built to keep organisms out. A better reading is an interface tuned over long co-evolution to be habitable by some organisms and not others.

Consider what those cells give up. As a keratinocyte moves towards the surface it degrades its own nucleus and organelles, ending as a protein-filled husk. It cannot

divide, repair itself, or respond. It also cannot be hijacked by a pathogen that needs host machinery to replicate, and it cannot accumulate ultraviolet damage in a genome it no longer has. The cell trades everything it could do for the one property the barrier needs, which is to be inert.

The surface is not uniform. Moist sites such as the armpit, oily sites such as the forehead and upper back, and dry exposed sites such as the forearm differ in acidity, lipid composition, moisture, and antimicrobial peptide concentration. Different communities occupy each. Bacteria tolerant of oil-rich, low-oxygen conditions dominate sebaceous ("greasy") sites; different types of bacteria dominate moist ones. Two sites on the same person can differ more from each other than the same site differs between two people.

That variation is not noise. It is a record of how many separate arrangements one organism's surface maintains at once.

What the barrier publishes

Two things follow that are worth stating plainly.

The first is that the skin does not assemble its microbiome. It sets conditions — acidity, available lipids, moisture, salt, antimicrobial peptides — and the community that establishes itself is whichever one happens to tolerate them. The barrier specifies an envelope and lets chemistry resolve the rest. This is the same procedure as

the tidal pool in chapter 3, on a different substrate and timescale: produce conditions, let differential persistence sort the outcome. Nothing selects in the sense of choosing.

The second is that the barrier is not a structure so much as a process. The stratum corneum is continuously rebuilt from beneath and continuously shed, turning over on a timescale of weeks. The outer layer of skin on a hand at this moment was not there a month ago and will not be there a month from now. It is a **dissipative structure** at the scale of an organ — its persistence depends entirely on not stopping. Interrupt the turnover and the result is not a stable barrier but one that fails at the first injury or test of its stability.

The relationship runs both ways, which is the part that resists the wall metaphor most. Animals raised without any microbial exposure develop differently: their skin immune environment does not mature normally, and specific commensal ("resident") bacteria have been shown to tune local immune responses independently of anything happening in the gut. The microbial community is not a population the barrier tolerates. It participates in building it.

How any of this is known

A brief detour into method, because the measurement technique is itself an instance of the book's subject.

Most of what is known about microbial communities comes from sequencing a single gene: the one encoding the **16S ribosomal RNA**, a structural component of the bacterial ribosome. It has an unusual property. Along its length, stretches that are nearly identical across the entire bacterial domain alternate with stretches that differ substantially between lineages.

Neither pattern is accidental. The conserved stretches sit in the parts of the molecule that do the work of translation (protein building), and translation is not optional — a lineage that mutates there does not continue. The variable stretches sit in parts where the sequence matters less, and over billions of years they have drifted freely.

So a single molecule carries two records at once: what selection could never permit to change, and what it never had reason to protect. The gene is a map of its own constraint.

That structure is what makes the measurement possible at all. The conserved regions give chemical handles that work on essentially any bacterium, known or unknown. The variable regions in between carry the differences that identify which bacterium it was. Amplify the gene from a sample, read the variable regions, compare against a reference, and the organisms present can be named without any of them ever being grown in culture — which matters, because most of them cannot be.

Carl Woese used this gene in 1977 to establish that archaea are a primary branch of life distinct from bacteria — what he later named a domain — which is among the larger revisions in the history of biological classification and was arrived at by reading a single molecule's record of its own constraints.

The method has real limits and the honesty of the chapter depends on naming them. Species carry different numbers of copies of the gene, which distorts abundance estimates. Resolution often stops at genus. The amplification step introduces biases. And it sees bacteria and archaea only — the fungi and viruses in the same communities require other approaches entirely.

The microbiome as an emergent system

A microbial community can collectively perform transformations no single member performs alone. Its composition affects nutrient availability, metabolite production, resistance to colonization by pathogens, and interaction with the immune system.

These effects depend on which organisms are present, in what proportions, doing what, in which environment. The same species can have different effects in different contexts; a function useful in one community may be useless or harmful in another.

The microbiome therefore cannot be understood as a list of species. What matters is the relationships —

among organisms, their genes, their chemical products, and the conditions they share.

This also complicates where an individual organism begins and ends. A human body has a clear physical boundary, but a great many of its functions are influenced by organisms living within and upon it.

The term **holobiont** is sometimes used for a host together with its associated microorganisms. It usefully emphasizes interdependence. It should not be taken to mean that host and microbes form a single evolutionary individual: the degree of integration, the stability of the association, and the extent of shared interest all vary, and in many cases the interests diverge outright.

The microbiome illustrates several recurring concepts:

- **Metacompartmentalization:** microbial communities occupy specialized environments within a larger organism.
- **Distributed labour:** different organisms contribute different metabolic and ecological functions.
- **Interdependence:** organisms exchange resources and influence one another's survival.
- **Emergence:** community-level properties arise through interactions among members.
- **Selection:** host conditions and microbial competition determine which populations persist.

- **Feedback:** microbial activity alters the environment, which alters the community.

It is a useful example because it shows that biological complexity does not always come from one organism becoming more internally specialized. It can also arise from relationships among organisms that remain distinct.

A microbiome in a tube

The same kind of community can be grown from a handful of mud. A **Winogradsky column**, named for the microbiologist Sergei Winogradsky, who devised it, is a clear tube filled with sediment enriched with a source of cellulose, such as leaf litter, and a source of sulfate, then left in the light. No organisms are added. Over the following weeks the mud sorts itself into layers, and the sorting can be watched through the wall.

The colours are chemistry. At the top there is oxygen, and it falls away with depth, so the mud below is **anoxic**, without oxygen. Decomposers break the cellulose into simpler carbon compounds that other microbes ferment. Down in the anoxic mud, bacteria respire sulfate in place of oxygen and release hydrogen sulfide, which combines with iron to form ferrous sulfide; the mud turns black, in one study within the first week. The sulfide diffuses upwards towards the oxygen, so each depth offers a different mix, and a different niche.

Bacteria that use hydrogen sulfide in photosynthesis live where light and sulfide overlap. Green sulfur bacteria typically form a green zone in the lower middle of the column, and purple bacteria a red-violet zone above it. Near the top, patches of orange appear, most likely iron oxides. On the inside of the tube, a biofilm grows that is less diverse than the mud behind it, with cyanobacteria among its commonest members; in columns kept in the dark, no visible surface community forms. The textbook bands are typical, not guaranteed: in one survey of columns, distinct green and red-violet zones did not appear.

Two results matter more than the colours. Organisms that are rare in the starting mud can become abundant once conditions suit them, so the column does not import a community; it selects one from what was already there. And no layer is independent of the others: what grows at each depth depends on what the layers around it produce and use up.

Here the author would go further. A column is not a holobiont in the sense used above, because there is no host. But the whole tube behaves a little like one: a self-contained system whose members live on one another's by-products, with a boundary, a division of labour, and conditions that its own residents keep remaking. That is where the analogy stops. There is no host whose interests the community serves, and no shared descent for selection to act on as a whole.

8
The origin and development of biological information

One of the central questions in the origin of life is how chemical systems began to retain and transmit information.

The **RNA world hypothesis** proposes that RNA may have played an early role in both information storage and catalysis. RNA can store sequence information, and some RNA molecules — **ribozymes** — can catalyse chemical reactions. A single class of molecule doing both jobs is an attractive way to connect chemical activity with heredity, because it removes the need for two separate inventions.

The RNA world remains a hypothesis about early evolution, not a complete or universally accepted account of life's origin. Other proposals involve peptides (small protein units), metabolic networks, mineral surfaces, or some combination of these. The framework here does not require any one of them to be correct; it identifies the organizational transition that every successful account has to explain.

Modern biological systems distribute the work across several components:

- DNA stores hereditary information.
- RNA transfers and regulates it.
- Proteins perform structural and catalytic work.
- Ribosomes translate genetic information into proteins.
- ATP moves usable chemical energy around the cell.

The shape of the code

Something worth examining closely sits in the middle of that list: the mapping between DNA sequence and protein, the **genetic code**.

The mechanics are simple enough to state. Three nucleotides make a **codon**. Four possible nucleotides in three positions gives $4 \times 4 \times 4$ combinations — 64 codons. Those 64 specify 20 standard amino acids plus a stop instruction. More codons than meanings means the code is **redundant** — several different codons specify the same amino acid.

The interesting part is that the redundancy is not spread evenly. It has a shape.

Most of the slack sits in the third position. For many amino acids, all four nucleotides in the third position of the codon specify the same amino acid. A copying error there frequently changes nothing about the resulting protein. Such changes are called synonymous,

and they are common. The flexibility of this position is sometimes described through the **wobble** in how transfer RNA pairs with the third base, a mechanism proposed by Crick in 1966.

The first two positions carry most of the meaning. A change there usually does change the amino acid. But not arbitrarily: codons that differ at the first position tend to specify amino acids with broadly similar chemical character. A substitution is more likely to swap one water-repelling residue for another than to replace it with something chemically opposite, and proteins frequently tolerate the former.

The control instructions are the least forgiving. Start and stop signals mark where a protein begins and ends. Disrupting them tends to produce more serious consequences than an internal substitution, because the error affects the whole product rather than one position in it. This is a matter of degree rather than an absolute: bacteria use several alternative start codons, read-through of stop signals occurs, and some organisms and organelles run variant codes in which a standard stop has been reassigned. The pattern holds; the exceptions are real and are part of why the code is studied as something that evolved rather than something fixed.

So errors are not treated uniformly. Where a mistake would be cheap, the code absorbs it. Where a mistake would be expensive, it does not.

Read as compression

That arrangement is what information compression looks like when it is working well: discard what can be discarded without losing the message, and protect what cannot.

The comparison is not merely impressionistic, and the standard code turns out to be unusually good at this. Freeland and Hurst compared it in 1998 against large numbers of randomly generated alternative codes, scoring each by how much damage a copying or reading error would do. Allowing for the fact that some kinds of error are more common than others, they found roughly one in a million random codes performed better than the real one. Their paper is titled accordingly.

The figure depends on which chemical property is weighted and which model of error is assumed, and it has been argued over since. One line of criticism is specific and substantial: when the differing frequencies with which organisms actually use particular codons are taken into account, the apparent optimization shrinks considerably. The general finding — that the standard code sits far out in the tail rather than in the middle — has held up. How far out is contested.

Three readings of that fact compete, and the differences matter:

- **Error minimization.** The code was shaped by selection, because lineages whose codes mistranslated and mutated more gracefully, without wrecking the proteins built from them, left more descendants. The structure is then an adaptation to the error rate of the machinery reading it.
- **Frozen accident.** Crick's 1968 proposal: the assignments were largely arbitrary, fixed very early, and became impossible to change afterwards, since altering what a codon means rewrites every protein in the organism simultaneously. Any error tolerance would be partly incidental.
- **Neutral origin.** The most interesting of the three for this book's purposes, and the one most easily missed. If the code grew by adding amino acids one at a time, with each new arrival assigned codons near those of a chemically similar amino acid already present, substantial error tolerance falls out of the growth process itself. No selection for robustness is required at any point. Models of this kind produce codes nearly as error-tolerant as the real one.

There are further proposals — direct chemical affinities between amino acids and their codons, or a code that developed alongside the biosynthetic pathways produ-

cing the amino acids. These are not mutually exclusive, and the current picture probably involves more than one. The question is open.

The third reading deserves the extra weight because of what it would mean. It describes a structure that looks optimized, performs as though optimized, and was never optimized for anything — the apparent design falling out of the order in which parts arrived. That possibility is worth holding onto, because it is the form every claim in this book is most at risk of taking.

What is not in dispute is the structure itself. Whatever produced it, the code distinguishes between positions where information can be lost safely and positions where it cannot, and it concentrates its tolerance in the former.

There is a resemblance here to error-correcting codes in engineering, and it is worth noticing without overstating it. An engineered code adds redundant symbols deliberately so that corruption can be detected and reversed. The genetic code does nothing of the kind: it tolerates error by how the mapping is arranged, not by adding check data, and it cannot detect or repair anything on its own. (Cells do have dedicated repair machinery, but that is separate from the code.) The two arrive at related behaviour by unrelated means — which is a resemblance, not a shared mechanism.

The sequence is not the whole message

One further point about DNA specifically. The molecule stores information in sequence, but it also has a shape, and the shape carries consequences. Strands bend and loop so that regions far apart in sequence can come into physical contact, and in many organisms this is part of how genes are switched on and off. The same sequence can therefore be read differently depending on how the molecule is folded and what is bound to it.

Information in biological systems is not held in one place or in one form. It is distributed across sequence, structure, chemical modification, and the state of the cell doing the reading.

Why the transition mattered

The arrival of a system capable of reliable heredity, variation, and reproduction was a major transition. Before it, chemical systems could persist. After it, they could accumulate. A pattern that is copied with occasional error, in a population where some copies do better than others, will change across generations without anything directing the change.

The exact route by which this transition occurred remains an open scientific question — and it is the largest remaining gap in the account this book is assembling. Chapters 3 through 5 describe chemistry that organizes, localizes, and sustains itself without any of it

being inherited. Everything after this chapter assumes heredity already exists — as, strictly, do chapters 6 and 7, which take up the theme of boundaries ahead of their place in the sequence. How the first reliably copied pattern arose is the step in between, and it is not yet known. The nearest approach so far is the 45-nucleotide RNA described in chapter 4, which can make both itself and its complement, slowly, inaccurately, and in separate reactions. Its discoverers still write that how the capacity for heredity first emerged is unknown.

9
Biological evolution and increasing organization

Once systems capable of heredity and reproduction existed, biological evolution could produce increasingly diverse forms of organization.

The components are few. Traits are inherited. Copying introduces variation through mutation. Some variants reproduce or survive better than others, so their frequency rises — **natural selection**. Chance alone shifts frequencies too, especially in small populations — **genetic drift** — and populations exchange material through **gene flow**. Ecological relationships change, which changes what counts as advantageous.

Nothing in that list specifies an outcome. Evolution has no predetermined destination and no tendency towards complexity. Simpler organisms are frequently the more successful ones by any measure that counts individuals or biomass, and complex structures are lost whenever maintaining them costs more than their function is worth. Whales lost their hind limbs. Cave fish

lose their eyes. Parasites routinely shed organ systems their hosts can supply instead.

What evolution does leave is a record — not only of what worked, but of the route taken to get there. Two kinds of evidence are worth examining closely, because they point in opposite directions and are both informative.

The same solutions, over and over

Unrelated lineages facing similar problems frequently arrive at similar answers.

Eyes are the standard illustration. Light-sensitive structures have arisen independently many times across the animal kingdom — a frequently cited survey put the number somewhere above 40, though counts vary with how strictly "independent" and "eye" are defined. Vertebrates built a camera eye with a single lens. Insects built a compound eye from many small units. Cephalopods built a camera eye too, separately.

Oxygen transport tells a similar story. Vertebrates use **haemoglobin**, which carries oxygen on an iron atom held in a porphyrin ring made of mostly carbon. Many molluscs and arthropods use **haemocyanin**, which uses copper instead, held directly by the protein rather than in a ring, and is not related to haemoglobin by descent. Some marine invertebrates use a third protein again. Different molecules, independently arrived at, doing the same reversible job.

Powered flight evolved in insects, in pterosaurs, in birds, and in bats — four times, from four different starting anatomies.

This is not coincidence, and it is not a tendency towards any particular form. The physics of focusing light, of binding oxygen reversibly, and of generating lift are the same everywhere, and they permit only a limited range of workable solutions. Evolution does not invent from nothing. It finds its way into the set of configurations that physics and chemistry leave open, and when the same problem recurs the same region of that set keeps getting found.

There is a complication worth stating, because it is more interesting than the clean version. The optical structures of eyes did arise separately, but the genetic machinery that initiates eye development is deeply shared — a regulatory gene family common to animals with eyes of all kinds is involved in building most of them. So "independent" describes the engineering, not the entire toolkit. Lineages reached similar destinations partly because they set out carrying similar equipment.

The marks of the route taken

Convergence shows what the constraints allow. The second kind of evidence shows what history imposes.

Consider the **recurrent laryngeal nerve**. It runs from the brain to the larynx, in the throat. It does not go there directly. It branches off the vagus nerve, descends

into the chest, loops beneath a major artery near the heart — the aortic arch on the left side, the subclavian artery on the right — and climbs back up to a destination a few centimetres from where it started. In a giraffe the same loop runs the length of the neck and back, about four and a half metres of nerve to span a gap of inches.

The explanation is developmental and historical. In fish, the equivalent nerve runs a short, direct course to its corresponding gill arch. As the ancestors of land vertebrates evolved, the heart and great vessels shifted backwards into the chest, and the nerve, hooked around a vessel, was carried along. Each incremental shift was small enough to tolerate. No step could undo the arrangement without disconnecting the nerve, and so the detour was retained and extended every time a neck got longer. The direct route was never restored: evolution worked around the rest of the body's layout, and the long way round cost too little to be selected away.

If easier, imagine an old city with old tiny roads. The tiny roads still lead where a driver needs to go, but the city grew around them, within limits set by the area's geography, and a drive across town can mean a weird "detour" a few blocks longer than walking directly there, free of one-way streets forcing odd turns. Old cities optimized around different conditions, and their streets are like the detour this nerve takes.

The mammalian middle ear is the same kind of story told forward. Two of the three small bones that transmit sound from the eardrum to the inner ear are modified from bones that formed the jaw joint in earlier vertebrates — the reptilian quadrate and articular became the mammalian incus and malleus.

This transition is unusually well documented, and it has an intermediate that settles the question of how such a change is even possible. Some early mammal relatives had **two working jaw joints at once**: the older one, between quadrate and articular, and a new one between different bones alongside it. With the load carried by the second joint, the first was free to be repurposed without the animal ever losing the ability to chew. There was no moment at which a jaw had to stop working in order for an ear to start.

The homology was proposed from embryology before the fossils were found, and the fossils then confirmed it.

Human development carries similar marks. Early vertebrate embryos — fish, birds, and mammals alike — develop a series of **pharyngeal arches**, ridges of tissue in the neck region. In fish these contribute to the supporting structures of gills. In mammals the same arches give rise to the jaw, parts of the middle ear, and structures of the larynx. The arches are homologous; what they become is not. (Human embryos do not have gills, and the older claim that an embryo replays its ancestry

is not correct. What is shared is the developmental starting material, not the sequence of adult forms.)

Some structures persist after their original role has diminished. Third molars — wisdom teeth — frequently lack room in a jaw that has become shorter than the one they evolved in. The coccyx is a set of fused vertebrae left from a tail; it no longer functions as one, though it is not inert, since pelvic floor muscles and ligaments attach to it.

The appendix is worth a paragraph of its own, because it used to be the standard example in lists like this one and it should no longer be. Evidence has accumulated that it has immune functions and may serve as a reservoir of gut microorganisms that helps repopulate the intestine after disruption. It has also evolved independently many times in mammals, which is difficult to square with its being useless. The lesson is not that the category of retained structures is empty. It is that "of unknown function" and "of no function" are different claims, and that the first has repeatedly been mistaken for the second.

Taken together these are the signature of a process that can only modify what is already present. It cannot set a structure aside and begin again, because every intermediate form has to be a working organism. It therefore reaches its solutions by the route that was available rather than the route that would be shortest, and the route stays visible in the result.

The tree is not only a tree

One more feature of biological history complicates the familiar branching picture.

Genetic material does not travel only from parent to offspring. It also moves sideways, between organisms that are not related, by several mechanisms — including viruses, which carry sequence from one host and deposit it in another. Among bacteria and archaea this **horizontal gene transfer** is common enough that a lineage's history is better described as a network than a branch.

Viruses are consequently poorly described as parasites alone. They are also a transport system, and material they have moved has repeatedly been put to use. The clearest mammalian example is **syncytin**, a gene derived from a retroviral envelope protein. Its original function was helping a virus fuse with a host cell membrane. In placental mammals it fuses cells to each other, forming a layer essential to the placenta — mice engineered without it fail to form a working placenta, and the embryos do not survive.

The striking part is how often this happened. Syncytins have been identified in primates, in several separate rodent groups, in carnivores, in ruminants, and in marsupials. They are not variants of one captured gene inherited from a common ancestor. They come from unrelated retroviruses, captured independently at differ-

ent times in different lineages, and co-opted for the same job each time. Placental mammals did not acquire this capability once. They acquired it repeatedly, from whatever virus was available, which is convergent evolution running on borrowed parts.

A substantial fraction of the human genome consists of sequence of viral origin, most of it inert, some of it in use.

Archaea matter to this picture too. They occupy conditions — extremes of temperature, salinity, pH — where other cells cannot persist, and their biochemistry differs from that of bacteria in fundamental ways including the chemistry of their membranes. They are not a marginal variation on bacteria. They are evidence that the early diversification of life explored genuinely different solutions.

All of this bears on how the base of the tree should be understood. There is strong evidence for a last universal common ancestor, in the form of features shared by everything alive: the same genetic code, the same ribosome, the same energy currency, the same handedness of key molecules. Reconstructions from those shared features describe something fairly specific — a small single-celled organism that would not look remarkable among modern bacteria.

Two complications sit on top of that. “Ancestor” in this context means a population rather than an individual, as it does everywhere else in evolutionary bio-

logy. And the early tree appears to have been genuinely web-like, with so much sideways transfer that the lines between branches are blurred near the base; one influential suggestion is that lineages only began behaving as distinct lineages once that exchange slowed. How far down the blurring goes, and whether it reaches the root, is not settled.

Here the author would go further: that early Earth, running an enormous number of parallel chemical experiments across many environments, is more likely to have produced self-sustaining systems in several places than in exactly one, and that what appears in retrospect as a single origin may be the merge point of a network that already had several. This is a minority position. The universal features listed above are real and do constrain how much independent origination the evidence will bear. It is offered as the author's reading, not as the current scientific consensus.

New levels of organization

Against a history this contingent, it is still true that biological organization has repeatedly found new levels.

Multicellularity arose independently in several lineages. Cells specialized, and specialization required coordination: signalling, regulation, shared developmental programmes, and mechanisms for resolving conflicts between what suits a cell and what suits the organism containing it. Tissues, organs, nervous sys-

tems, and complex social behaviour followed in various lineages at various times.

Each of these is the pattern from chapter 6 appearing again — containment, interaction, specialization, interdependence, coordination — now operating on cells rather than molecules, and over evolutionary rather than chemical timescales.

The capability of the resulting system depends less on what its individual components can do than on how they are related. That is the thread the later chapters follow, as it runs through nervous systems, language, and culture.

10
Spirals, helices, and recurring geometry

Spirals and helices turn up everywhere once noticed.

DNA is a double helix. Many proteins contain helical segments. Bacteria swim by rotating a helical filament. Nautilus shells, ram horns, and elephant tusks grow as spirals. Sunflower heads and pine cones are built from interlocking spiral families. Climbing plants coil. Water drains in a vortex, hurricanes rotate, and some galaxies carry spiral arms. A whole class of animals is classified partly by the spiral pattern their earliest cell divisions follow. And the same shapes are in the garage: a garden hose coiled for the winter, a rope, the line on a fishing reel — ordinary objects that, as a later section shows, are solving the problem a chromosome has.

The usual response to a list like that is either to treat it as a coincidence or to treat it as a sign of something deep. Neither is right, and the reason why is more interesting than either.

Five ways to make a spiral

A spiral is not one thing with one cause. It is the output of several unrelated processes, each of which arrives there for its own reasons. The recurrence is real; the explanation is plural.

Growth proportional to size. A structure that adds material at its edge, with each increment scaled to the current size and set at a constant angle, cannot grow as anything but a logarithmic spiral. A nautilus adds shell at the opening; the new chamber is larger than the last by a fixed ratio; the shape follows by necessity. Snails grow in a similar manner. The animal is not executing a spiral. It is adding to its edge at a constant proportional rate, and a spiral is what that looks like from outside. D'Arcy Wentworth Thompson worked this out in detail in 1917, and it remains the clearest demonstration in the subject: a growth rule produces a form without the form being specified anywhere.

Rotation combined with advance. Anything that turns while moving forward traces a helix. DNA's double helix comes from stacking flat base pairs with a slight consistent twist. A bacterial flagellum is a helical filament driven by a rotary motor. A tendril coils because its two sides change length by different amounts. Same shape, three mechanisms with nothing in common.

Sequential packing under constraint. This is the one that looks most like design and is least like it. A sunflower head produces new structures one at a time at its growing centre, and each appears in the largest gap available. Repeat that rule and the angle between successive units converges on about 137.5 degrees, which generates the interlocking spiral families and the Fibonacci counts that go with them. The decisive evidence is that the pattern is not biological at all: Douady and Couder produced the same spirals in 1992 using drops of magnetized fluid repelling each other on a dish. No genes, no organism, same geometry. The pattern comes from the packing problem, not from the thing solving it.

Rotation with inward or outward flow. Material converging while rotating conserves angular momentum and spins faster as it draws in. Whirlpools, hurricanes, and accretion around a massive body all follow. (Water leaving a household drain is not an instance of this at planetary scale — the direction is set by the basin's shape and whatever motion the water already had, not by the Earth's rotation.)

A wave curling around a free end. The least familiar of the five, and the one that makes a turning spiral with no material rotation at all. In a medium where each point can be triggered by its neighbours and then needs time to recover before it can be triggered again, a travelling wave front that is broken — with a loose end rather than a closed ring — will pivot around that end. The tip

turns fastest because it has the least recovered tissue ahead of it, and the result is a rotating spiral wave that can persist for hours. Nothing is moving in a spiral; the pattern is a sequence of excitations passing through a stationary medium. The Belousov–Zhabotinsky reaction in chapter 4 produces exactly this in a dish, and the same mechanism generates certain dangerous cardiac arrhythmias, where a wave of electrical excitation circulates in heart tissue instead of sweeping across it once.

Five conditions, five mechanisms, one family of shapes.

What the recurrence shows

The honest reading is this: a spiral is what extension looks like when something must keep a fixed relationship to where it has already been. Spider webs and spiral staircases show it too.

That constraint is the common factor, and it is a real one. A structure that grows only at its edge cannot revisit what it has already built. A filament that turns while advancing cannot occupy the space it just left. A head packing new units cannot displace the old ones. In each case the system must extend without discarding its own history, and the set of shapes that permits is small. The spiral is less a thing nature favours than a thing geometry leaves available.

Stated that way the claim is strong, and it is also the kind of claim that can be wrong. It makes a prediction:

systems that are free to rearrange themselves wholesale, or that have settled into a stable configuration, should not show this geometry. Broadly they do not. A mountain is a load-bearing compromise and is blocky or conical. A planet large enough to pull itself into equilibrium is a sphere. A crystal repeats a fixed cell. These are shapes of having arrived rather than shapes of still going.

Two cautions keep this from running away.

The first is that similar form does not imply shared mechanism. This is the point the whole book keeps returning to, and this chapter is where the temptation to forget it is strongest. A nautilus shell and a spiral galaxy look alike in photographs and have nothing whatever in common underneath. Galaxy arms are not material structures that wind up over time; they are better understood as waves of compression passing through the disc, with stars moving in and out of them. Putting those two images side by side invites an inference that the physics does not support.

The second is that "spiral" is doing loose work across the list. A logarithmic spiral, a helix, a phyllotactic lattice, and a vortex are distinct objects. Grouping them under one word is a convenience, and conveniences of that kind are where arguments go wrong.

Coiled for storage

The constraint behind every spiral in this chapter — extension that cannot revisit what it has already laid down — has a working counterpart in how long, thin things are stored. A rope, a cable, a fishing line, a garden hose, and a chromosome share a problem. Each is long in one dimension, has to be kept in a small space without tangling, and has to come back out in order. The answer, every time, is to coil it or wrap around something continuously.

Coiling has a cost that anyone who has wound a garden hose by hand has met. A strand coiled in loops laid the same way each time picks up a full twist per loop, and when it is pulled straight again the twist has to go somewhere, which is why a hand-coiled hose kinks as it comes off the pile. The accounting is exact. For a strand whose ends cannot turn freely, a quantity called the linking number stays fixed, and it is the sum of two others: twist, how far the strand turns about its own axis, and writhe, how far that axis coils through space. Coil the strand and writhe goes up, so twist must come down to compensate, and the reverse when it is uncoiled. A reel that turns its drum lays a hose or a line down without this problem; a fixed-spool spinning reel, which wraps line around a stationary spool, twists the line with every cast and retrieve, and line twist is the familiar complaint about them. Cable technicians coil

over and under — twisting one loop, untwisting the next — so the twists cancel and the cable lies flat when it is thrown out. Rope puts the same accounting to structural work: fibres are twisted into yarn one way, yarns into strands the other, and strands into rope the first way again, each counter-twist locking the layer beneath it. What appears to be three-strand laid rope, about twenty-one thousand years old, was found in the cave at Lascaux.

DNA has every one of these problems, at a scale that makes them severe. Circular DNA from a virus was found twisted on itself in 1965, and the writhing number was defined in 1971 in response to the questions that raised. In the cells of plants and animals the double helix is wrapped around cores of protein, 146 base pairs at a time and about thirty million times in each human cell, and the wrapped strand is folded again on itself. And it is stored to be read in order. Reading means opening the helix, and a machine moving along a twisted strand piles extra twist up ahead of it and leaves a deficit behind — the spinning reel's problem exactly. Cells deal with it using enzymes that cut the DNA, let it turn or pass through itself, and join it again: topoisomerases, which do for a chromosome what a swivel does for a fishing line or an anchor chain.

Here the author would go further. Rope, cable, line, and chromosome are one kind of object: a one-dimensional sequence that has to be packed into three dimensions and

later unpacked in the order it went in. A coil is the shape that does both. It is the stored form of something meant to be paid out, and its spiral is the record of how it was laid down — this chapter's principle, extension that keeps its own history, put to practical use. The resemblance holds in the mathematics, which is why one piece of accounting covers a hose and a chromosome. Whether it is more than that is the question chapter 17 asks of every resemblance in this book.

Spirals people make

People have been making spirals for about as long as they have been making marks that last. The passage tomb at Newgrange in Ireland, built around 3100 BCE, carries carvings that fall into a small number of recurring families — some curved, among them circles, arcs, and spirals, and some straight-edged, among them chevrons, parallel lines, and lozenges (diamond shapes). Its best-known stones set the two kinds side by side. The entrance stone carries spirals with lozenges beside them. On the stone directly opposite, at the back of the mound, spirals occupy the upper part of one panel and a lattice of lozenges fills the space beneath. Inside, a triple spiral is carved on the wall of the chamber that the rising sun reaches once a year, at the winter solstice. What any of it meant to the people who carved it cannot be recovered.

Here the author would go further. The pairing reads as a record of the two ways a bounded space gets filled. A spiral extends from a centre, adding at its edge and never revisiting what it has already laid down, as the growing structures earlier in this chapter do — the shape of something still going. A lattice of lozenges repeats one cell until the edge of the stone stops it, as a crystal does — the shape this chapter called having arrived. Set together on a surface with fixed edges, they are growth and tiling, the organic and the crystalline, each filling the room it is given. Nothing is known about whether the carvers saw it that way. The resemblance is in the geometry rather than in any recorded meaning, and it is offered as a reading, under the same caution as everything else in this chapter.

Spirals were carved and painted independently in many places that had no contact with Ireland or with one another. More than 1,400 petroglyphs at Las Plazuelas in central Mexico are mainly spirals, dot figures, and carved models of buildings, and spirals are among the Nazca lines in the Peruvian desert. The Māori koru is a spiral modelled on the unfurling frond of a fern — a growth spiral copied from a growing thing. In the Greater Antilles, Puerto Rico among them, the Taíno called the great storm *juracán*, the source of the word hurricane, and the Cuban scholar Fernando Ortiz argued in 1947 that Taíno images of the storm used a rotating figure, in its most abstract form a spiral: the storm drawn as the rotation it is. At Chaco Canyon in New

Mexico, spirals pecked into a cliff on Fajada Butte are crossed by blades of sunlight, falling between leaning slabs of rock, at the solstices and equinoxes — a solar marker built around a spiral, like Newgrange. Whether the slabs were set there or were a natural rockfall the carvers made use of is disputed; either way, the spirals appear to have been placed to be lit.

There are several candidate answers to why the same mark turns up in so many unconnected places. Some spirals are copied from things that spiral, like the fern and the storm. Some are tied to the sky. And one proposal locates the source inside the observer. Geometric hallucinations fall into four recurring families, two of which are lattices and spirals — the pair carved together at Newgrange — and mathematical models of the wiring between the eye and the visual cortex reproduce those families from the structure of the cortex itself. Archaeologists have argued that Irish passage-tomb art, and much older cave art, draws on imagery of this kind. That argument is contested. The answers are not exclusive, none is established for any particular stone, and a shared form says nothing about a shared meaning. What the recurrence does show is this chapter's argument arriving from another direction: the spiral is available to anything that extends from a centre, a hand moving a tool included.

A comment from the author, not a finding: the potter's wheel may be the plainest case of a person working with a

spiral rather than drawing one. The wheel supplies the turning; the potter's hands supply the advance, drawing the clay up and out as it spins, and rotation combined with advance is this chapter's second way of making a spiral. The pot is the record of that path.

The most famous painted spiral has a better-documented history, and it turns the first caution above into a worked case. Vincent van Gogh painted *The Starry Night* in June 1889, from the window of his room in the asylum at Saint-Rémy-de-Provence, where he had admitted himself the month before. The village is invented, but much of the sky is not. Venus, which he described in a letter as the morning star, was visible before dawn that spring at close to its greatest brightness, and it is the brightest point in the painting; the moon, by contrast, is shown in a phase it was not in. The great swirl across the centre is the element nobody has been able to place. An art historian proposed that it echoes the drawings of spiral nebulae — galaxies, as they were later understood to be — reproduced in the popular astronomy books of the day, and an astronomer studying the painting independently agreed, tracing the image to Lord Rosse's 1845 drawing of the Whirlpool galaxy, while also suggesting the mistral, the wind that dominated van Gogh's months in Provence. Both readings are plausible. Neither can be checked.

In 2024 a team of fluid dynamicists tried measuring the painting instead of interpreting it. Treating the

brightness of the paint as a stand-in for a quantity carried by a flow, they found that its spectrum — how much variation there is at each scale, from the largest swirl to the smallest — falls off with the same slope that Andrei Kolmogorov's 1941 theory predicts for fully developed turbulence, and concluded that van Gogh must have observed real turbulent flow with unusual care. An earlier study had reported a related match in the statistics of brightness across several of his late paintings.

The objections came from turbulence specialists, and they are the instructive part. The theory being applied describes a physical quantity carried by a moving fluid; a painting has no such quantity, and nothing guarantees that its brightness behaves like one. Then a second group ran the same analysis on a portrait by Degas of a woman seated beside a vase of flowers — no sky, no wind, nothing turbulent anywhere in it — and obtained the same slope. A test that a vase of flowers also passes cannot be what identifies turbulence.

That is the first caution in its most useful form. This time the resemblance between a painting and a turbulent sky was not eyeballed. It was measured, published, and still did not establish a shared mechanism, because the measurement turned out to be one that unrelated images pass as well. The painting does not need the claim. What van Gogh put on the canvas is a sky that looks as if it is moving, made by a hand moving paint.

A comment from the author, not a finding: perhaps van Gogh was watching a turbulent river and was inspired by its texture, or, more literally, saw turbulence in fast-moving clouds turn the sky wispy and spiral-like. The measurement cannot settle that, so it stays a guess. The alternative is to accept that the spirals suit the painting, which its popularity seems to bear out.

Becoming rather than being

What follows is the author's reading rather than an established result.

Take the observation seriously and it suggests a way of sorting systems. On one side, systems at or near equilibrium: settled, symmetric, finished, their shape determined by the balance of forces acting on them now. On the other, systems still running — growing, oscillating, circulating, recursing — whose shape records the path they took to get here as well as the forces acting on them.

The second group is where the spirals are. Not because a spiral is optimal, and not because anything prefers it, but because a system that is still extending and cannot undo its own past has a restricted set of forms available, and this family is most of that set.

This connects forward to chapter 13 more directly than it may appear. A mind modelling itself modelling never reaches a resting state; each pass takes the result of the last as its input. That is recursion rather than

rotation, and calling it a spiral is a metaphor rather than a measurement. But it has the shape of the condition described above: continuation that carries its own history forward and cannot start over. Chapter 11 takes up the same structure in time rather than space.

Where the reading stops

A note on the limits of that reading, since the limits are instructive. It is tempting to extend the argument to the hair whorl at the crown of the head — a visible spiral, sitting directly above the part of the brain most associated with deliberate thought. The temptation should be refused. Hair whorls arise from the orientation of follicles during skin development and have no relationship to the tissue underneath; the apparent significance is entirely an artefact of where the observer is standing. The pull of that connection is strong, it is exactly the pull chapter 13 describes, and noticing it without acting on it is the only defence available.

Coda: a planet that hums

One further place the geometry of oscillation shows up at scale, offered as a fact rather than an argument.

The gap between Earth's surface and the lower ionosphere forms a cavity, and lightning — roughly fifty strikes somewhere on the planet every second — rings it continuously. The cavity has resonant frequencies, the lowest around 7.8 Hz, and they are measurable with the

right instruments anywhere on Earth. These are the Schumann resonances, predicted in 1952 and confirmed experimentally within the decade.

Separately, Earth's liquid core generates a magnetic field, and that field is shaped from outside by the solar wind — compressed on the sunward side, drawn into a long tail on the other, flexing continuously as solar activity rises and falls.

Neither of these is mysterious and neither is doing anything for life as far as is known. They are included here because they are true and because they extend the chapter's subject to planetary scale: a system under continuous energy input, oscillating rather than settling, with a measurable geometry that exists only because it never stops being driven.

Nothing further is claimed for them. Speculation about these signals interacting with biology, technology, or anything else is not supported by evidence and is not part of this account.

11
Time as cycles plus accumulation

Many natural processes are cyclical.

Day and night, seasons, tides, biological rhythms, planetary orbits, and many chemical oscillations repeat over time. But cycles rarely repeat into identical conditions. Each one occurs in a system that has been changed by the ones before it.

A useful conceptual image is a series of circles accumulating along another axis, producing a spiral:

- The circular motion represents recurrence.
- The progression along the axis represents elapsed time.
- The expanding or shifting path represents accumulated change.

This is not a claim that physical time has a spiral geometry. It is a way of holding two things at once — that a process repeats, and that repeating it changes what it repeats into.

Cycles that keep their own records

The combination is not abstract. Several natural systems produce a physical record of it, which is why so much of what is known about the deep past comes from reading them.

A tree adds a growth ring each year. The cycle is annual and the same each time; the ring is not. Its width records that year's water, temperature, and competition, and because the rings accumulate rather than overwrite, a trunk holds an ordered sequence reaching back as far as the tree does. Overlapping records from living trees, dead timber, and archaeological wood extend the sequence back thousands of years.

Ice does the same with snowfall. Each year's accumulation is compressed by the next, and trapped air bubbles preserve samples of the atmosphere at the moment the snow closed over. The longest continuous Antarctic record reaches back about 800,000 years, across eight glacial cycles, and carries a direct record of past carbon dioxide — not inferred from a proxy, but measured from the air itself.

Lake sediments layer seasonally. Some corals and shells band with the tides. In each case a cycle that could in principle repeat unchanged instead leaves a residue, and the residue is what makes history legible.

The general form is worth naming. A system whose current state depends not only on present conditions

but on the path it took to reach them exhibits **hysteresis**. The term comes from magnetism, where a material's response depends on what was done to it previously, and the same structure appears wherever a cycle leaves something behind.

The same structure in living and cultural systems

Evolution is generational, and each generation starts from the population the last one produced. Learning is repetition that alters what repetition does. Development runs through stages that are similar across individuals and never identical, because each stage builds on the particular outcome of the one before. Cultural transmission passes practices on, and what is passed on has been reshaped by every previous passing.

None of these is literally a spiral, and applying the word to them should be read as the convenience it is. What they share is structural: a process that recurs, inside a system that retains the consequences of its earlier recurrences.

Chapter 10 described this in space — extension that cannot revisit its own past. This is the same constraint operating in time. A cycle that left nothing behind would be a closed loop, and a system made of closed loops would have no history to read.

An aside on cycles at the largest scale

What follows is the most speculative material in the book. It is included because the pattern of this chapter has been proposed at cosmological scale too, and because the arguments are interesting. None of it is established, and the chapter stands without it.

The question of whether the universe had a beginning or a previous state has several live answers, and at least three of them take a cyclic shape.

Loop quantum cosmology proposes that the Big Bang was not a singularity but a bounce: a prior contracting universe reaching a minimum volume at which quantum effects make gravity repulsive, and rebounding. On this account the beginning was a transition rather than an origin.

Conformal cyclic cosmology, developed by Roger Penrose and set out for general readers in *Cycles of Time* (2010), proposes a sequence of aeons in which the remote future of one — expanded, cooled, with its black holes evaporated and little left but radiation — is geometrically equivalent to the hot dense beginning of the next. Penrose and collaborators argued in 2020 that faint circular features in the cosmic microwave background are traces of black hole evaporation in a previous aeon. Other groups have disputed the analysis, arguing the features are consistent with chance in random data, and the claim is not accepted.

Cosmological natural selection, proposed by Lee Smolin in 1992, is the most structurally interesting of the three for this book's purposes, because it applies the vocabulary of chapter 9 to physics itself. The suggestion is that black holes produce new regions of spacetime with slightly different physical constants, that universes producing more black holes therefore leave more descendants, and that the constants observed here are the ones that happen to be good at making black holes.

Smolin's proposal deserves attention for a reason unrelated to whether it is true: a falsifiable prediction was derived from it. Combined with calculations of how matter behaves in the cores of neutron stars, published in 1994, maximizing black hole production implies an upper limit on neutron star mass, which Smolin put at about 1.6 times the mass of the Sun. He stated the stakes plainly in 2004: a neutron star heavier than about 1.5 solar masses would be troubling for the theory, and one of 2.5 would certainly refute it.

A pulsar of nearly two solar masses was measured in 2010; one at roughly 2.08 followed, and another at around 2.35 after that. The limit he committed to has been exceeded several times. The higher figure he named as certain refutation has not yet been reached.

Smolin has argued the proposal can be rescued; others treat it as refuted. Either way it did the thing this book's own weakest material does not do, which is put a number where it could be checked, so that the result

could come back wrong — and it did. Chapter 17 returns to why that is harder than it sounds.

The attraction of all three is obvious, and it is worth being explicit about why that is a reason for caution rather than confidence. A reader who has just been shown cycles that accumulate at every other scale will find a cosmic version satisfying, and that satisfaction is not evidence. The physics here is contested among specialists, and the pattern-matching that makes it appealing is no substitute for their understanding. It is included as an open question, and chapter 17 says more about why questions of this kind are where a framework like this one is most likely to mislead.

12
Life as a category

The definition of life is less straightforward than it first appears, and the difficulty is more interesting than a definitional quibble.

Life is generally recognized through a cluster of characteristics: metabolism, energy use, organization, growth, response to the environment, reproduction, heredity, and evolution. Not every living thing displays all of them in the same way, and various non-living systems display some of them convincingly.

Things that do some of it

Fire metabolizes, in a loose but not empty sense. It consumes fuel, releases energy, maintains itself as long as supply lasts, grows, responds to its environment, and spreads from place to place in a way that resembles reproduction. What it has no trace of is heredity. One fire does not pass traits to the next, so nothing accumulates and nothing evolves. Fire is the cleanest demonstration that the metabolic properties on their own are not sufficient.

Crystals grow by adding material in an ordered arrangement, and they propagate structure: a seed crystal placed in a solution produces more crystals of the same form. Pattern is being copied. But the pattern admits almost no variation that could be inherited and selected, so again nothing accumulates.

Stalagmites grow, are organized by physical and chemical conditions, and take on forms that depend on their local history. They do not reproduce.

Prions are proteins that propagate a misfolded shape by inducing the same misfolding in normal copies of the same protein. Structural information is copied from one molecule to another with no nucleic acid involved at all. Whether this counts as heredity depends entirely on how the word is defined.

Viruses carry genetic information, mutate, evolve under selection, and are unambiguously subject to the processes this book describes. They also cannot metabolize or reproduce without commandeering a host cell. Their status has been argued for a century without resolution.

The viral case has become harder rather than easier. Giant viruses discovered in the last two decades have genomes larger than those of some free-living bacteria and encode components of the protein-synthesis machinery that viruses were supposed to lack entirely. Meanwhile various bacteria — including those causing typhus and chlamydia — have lost so much of their own

metabolism that they cannot reproduce outside a host cell either. The property once used to separate viruses from organisms turns out to be held by neither group exclusively.

These cases demonstrate why it matters to distinguish growth, energy use, information, replication, heredity, metabolism, and evolution rather than treating them as a package. They come apart. Different systems hold different subsets.

Why no definition has settled the matter

Attempts at a formal definition exist in quantity. One compilation counts over a hundred and twenty distinct published definitions of life, which is itself the most informative fact in this section: a concept with one definition has been pinned down, and a concept with a hundred and twenty has not.

The most widely used working version — adopted by a NASA committee needing something operational for the purpose of looking for life elsewhere, and originally suggested by Carl Sagan — describes life as a self-sustaining chemical system capable of Darwinian evolution. It is compact and does real work.

It also has a well-known problem, and it is the obvious one. An individual organism that cannot reproduce — through sterility, age, or being the only one of its kind — is not itself capable of Darwinian evolution, though nobody would call it non-living. The definition describes

populations and lineages better than individuals. Adjusting it to cover individuals tends to let in things it was meant to exclude.

This pattern is general. Every proposed definition either rules out something uncontroversially alive or admits something uncontroversially not, and the usual response is a further clause, which produces a definition that is accurate and no longer says anything. A hundred and twenty attempts have not converged, and the failure to converge may not be a failure of effort.

The boundary is not missing information

It is worth being precise about what kind of problem this is, because the obvious reading is wrong.

Consider a heap of sand. Removing one grain never turns a heap into a non-heap. Yet a single grain is not a heap. There is no particular grain at which the transition occurs, and this is not because the matter has been inadequately investigated. The concept has fuzzy edges, and no amount of further study of sand will sharpen them.

"Life" appears to be the same kind of concept. It picks out a cluster of properties that usually travel together, and it works well in the ordinary case precisely because they usually do. It gives no answer at the edges because it was never the kind of thing that had one.

The consequence matters for everything in chapters 3 through 5 and 8. **"When did life begin" is mal-**

formed in the same way "when did it become a heap" is malformed. Those chapters describe properties accumulating one at a time — chemistry that persists, boundaries that localize, networks that sustain themselves, structures that copy — each step small, each step chemical, no step requiring anything the preceding one lacked. The question expects a moment. There is no moment, and the absence is not a hole in the account.

This should not be mistaken for the claim that there is no problem. There is a large one, named in chapter 8 and again in chapter 17: how a chemical system first came to copy a pattern reliably enough for selection to accumulate is not known. What dissolves is the expectation of a single threshold event with a single explanation. What remains is a sequence of transitions, at least one of which is genuinely unexplained. That is a different shape of ignorance, and a more tractable one.

The sample problem

One further limit, more serious than the definitional one.

Every characteristic in the opening list was derived from examples of a single kind: terrestrial, carbon-based, water-dependent, cellular, descended from a common ancestor. The definition of life is a generalization from one case.

A category built from one example cannot be known to be general. There may be organized, energy-driven

systems that fail the familiar criteria and are nonetheless the kind of thing the category was reaching for. There may equally be none. The possibility is open, and openness is not evidence.

What can be said is narrow: it is possible to measure familiar biological life with great precision while still being unsure how widely the category should extend. Those are different competencies, and the first does not supply the second.

13
Cognition: information becomes representation

Biological systems developed increasingly sophisticated ways of sensing and responding to their environments.

Nervous systems allow organisms to process signals, coordinate behaviour, and respond to changing conditions. In more complex brains, information can be integrated into representations that support memory, prediction, categorization, and decision-making.

Cognition can be understood partly as a process of information compression.

What a brain throws away

The compression claim is easy to state and worth making concrete, because the quantity discarded is the surprising part.

The discarding starts before anything reaches the brain, and it can be counted. A human retina holds something like ninety-six million photoreceptors. The optic nerve carrying their output to the brain contains roughly one million fibres. Around a hundred receptors

feed each one, and the reduction is not uniform — at the centre of vision each receptor has an output fibre to itself, and may have two, while towards the edge a single fibre serves hundreds.

So the eye is not a camera sending pictures inward. It is a device that throws away about ninety-nine per cent of what it collects before transmission, and what survives is already processed into edges, motion, contrast boundaries, and change rather than a faithful record of light levels. By the time anything reaches awareness it has passed through several further stages of selection and summary. The experience of seeing a room does not involve holding the room in mind. It involves holding a model adequate for acting in the room.

The model is built from regularities. A brain identifies what recurs, treats recurrences as instances of a category, and stores the category rather than the instances. This is why a familiar object is recognized from an angle never seen before, and also why an eyewitness account degrades towards the typical case.

Cognitive processes include:

- Pattern recognition.
- Categorization.
- Association.
- Memory.
- Prediction.
- Learning.

- Attention.
- Abstraction.

Every item on that list discards something. Attention discards most of the available input. Categorization discards the differences between members of a category. Memory discards detail and reconstructs it on demand, which is why recollection is more like rebuilding than like retrieval.

The payoff is prediction. A system holding a compressed model of its environment can anticipate what has not happened yet, and act before it does. An organism responding only to present stimuli is always late.

This is the same trade that appeared in chapter 8, running on different hardware. The genetic code tolerates error where error is cheap and resists it where error is expensive. A nervous system discards detail where detail does not change what to do, and preserves it where it does. Neither system was designed to compress. Both ended up compressing, because a system that keeps everything cannot act in time.

The loop that models itself

A model of the environment is useful. A model that includes the modeller is a different kind of thing.

An organism that represents its own body, its own position, and eventually its own representing has begun to do something recursive. Douglas Hofstadter gave this

the name **strange loop**: a system that, by moving through its own levels of description, arrives back where it started. His argument, developed in *Gödel, Escher, Bach* and more directly in *I Am a Strange Loop*, is that the sense of being a unified self is what such recursion feels like from inside. Not a thing located somewhere in the brain. A pattern the brain produces by modelling itself modelling.

Marvin Minsky's *The Society of Mind* approaches the same conclusion from the other direction. On his account the mind was never unified to begin with. It is a large number of specialized processes — handling vision, sound, balance, memory, planning, threat — none of which individually contains a self, and which collectively generate the appearance of one through their interaction.

A murmuration of starlings is a useful comparison, provided it is held as a comparison. The flock turns as a unit and appears to be directed. No bird is directing it. Each responds to a handful of neighbours, and the coordinated shape is what those local responses produce at scale. The pattern is entirely real. A central decider behind the pattern is not. The resemblance to a mind is structural rather than mechanistic — neurons and starlings have little in common beyond both being many things interacting locally — but it makes the point that unified-looking behaviour does not require a unifier.

This recursion also marks a threshold worth naming, because it is the one the rest of the book runs through:

- Self-replicating chemistry carries a minimal kind of self-reference: *make more of this.*
- Genetic systems extend it: *encode the pattern for making more patterns.*
- Nervous systems model the world.
- Some nervous systems model their own modelling.
- Language externalizes that model, so another mind can receive it.
- Culture accumulates externalized models across generations.

Each step reuses what the previous one built. None of them required the next one to follow.

One caution. None of this explains why information processing is accompanied by experience — why there is something it is like to be a system modelling itself. Hofstadter and Minsky describe an architecture. The question of felt experience is not answered by describing it, and chapter 17 returns to this as one of the places the framework stops.

Good and bad before anything else

Underneath representation sits something older and simpler.

Organisms assign value. Approach this, avoid that. A bacterium swimming up a nutrient gradient is doing a

rudimentary version; a plant growing towards light is doing another. These are not decisions, and nothing needs to be experienced for them to work. But they are the beginning of a system that sorts the world into better and worse.

As environments and bodies became more complicated, so did the sorting. More signals, more possible responses, longer time horizons, and eventually states that integrate many inputs at once — the layered and often contradictory things humans call emotions.

There is a tempting move here, and it should be resisted. It is tempting to say that consciousness *is* this valuation system, scaled up and nothing more. That claim does real work: it makes the capacity continuous rather than a sudden arrival, which fits everything else in this book, and it explains why the machinery of value is so deeply conserved. But it does not touch the question of experience raised above, and treating it as though it does would be the kind of overreach this book is trying to avoid.

What can be said is narrower and still useful. The sophistication of information processing varies continuously across living systems. Drawing a line somewhere on that continuum and calling everything above it conscious is a decision about where to put a word. It is not the discovery of a boundary in nature.

A fugue is an exercise

One concrete case of how a brain might be reshaped by what it processes.

Following a Bach fugue means tracking several melodic lines at once. Each is independent, each is mathematically related to the others, and the relationships — inversion, transposition, one voice entering against another already in progress — have to be held simultaneously rather than in sequence. The listener is not receiving a melody. The listener is running several, in parallel, and detecting their relations.

That is a demanding exercise, and the theory is that demanding exercises change the systems that perform them. The body offers an undisputed version: a guitarist's fingertips harden into calluses where the strings press, and nobody suspects guitarists of being born with hard fingertips. A brain is harder to read, because most of the evidence compares different people rather than the same person before and after. Trained musicians show measurable differences in auditory and motor regions relative to non-musicians, and the differences scale with training.

The direction of causation is genuinely unsettled, and the honest version has to say so: people who take up music may differ beforehand in ways that predispose them both to continue and to show those differences. Longitudinal work has found changes that follow train-

ing, which argues for real effect, but the size of that effect and how far it generalizes beyond music are both disputed.

What survives the dispute is the structural point. A fugue is a piece of external structure whose internal organization requires parallel, recursive processing to follow. Engaging with it exercises exactly that. Whether or not the benefit generalizes, the arrangement is worth noticing: a cultural artefact built to be complex in a specific way, which exercises the capacity needed to handle it and, if the theory holds, builds it.

The same capacity, pointed the wrong way

These processes are powerful and they are not reliable.

The faculty that finds genuine structure also finds structure that is not there. Humans are unusually good at transferring patterns across domains — noticing that two unlike systems seem to work alike — and this supports analogy, invention, scientific insight, and storytelling. It also produces confident connections between things that have nothing to do with each other.

The capacity has no reliable internal signal distinguishing the two cases. In experiments, solutions that arrive with a feeling of sudden fit are more often right than solutions worked out step by step, and the feeling is stronger for right ones; but wrong ones can bring it too, and a feeling of fit borrowed from something unrelated can make a false statement seem truer.

This is the problem the Preface describes from the inside, and it is the reason this book keeps marking which claims are established, which are contested, and which belong to its author. The marking is not modesty. It is the only available correction for a faculty that cannot reliably audit itself.

14
Language as cultural protochemistry

Language is a system in which small units combine into larger ones.

Sounds combine into words. Words combine into phrases and sentences. Grammar constrains how elements may be arranged, and meaning arises from the relationships among them together with shared social convention.

Language changes through use, variation, selection, retention, and transmission. New words and expressions appear; some are adopted widely and others disappear. Pronunciations shift, grammatical structures change, and meanings move.

The phrase **language as protochemistry** describes the combinatorial part of this: small reusable units, constrained ways of joining them, and complexity built from both. It is an analogy and not a claim that language operates through chemical reactions. What the two have in common is a combinatorial structure with

rules about what may attach to what. The mechanisms have nothing else in common at all.

Some recurring sounds across unrelated languages reflect shared anatomy rather than shared history. Words resembling *mama* and *dada* recur widely in part because the consonants involved are among the earliest an infant vocal tract produces reliably, and because caregivers are present to interpret those sounds as referring to them. Similar forms can arise from common constraints as easily as from common descent.

Does vocabulary shape thought?

This is the question the chapter has to handle carefully, because it is the one where popular accounts and the evidence diverge most.

The famous example, and why it is not used here

The standard illustration is that Inuit languages have an extraordinary number of words for snow, and that this lets their speakers perceive distinctions other speakers cannot.

The claim as usually told is not reliable, and the chain by which it became unreliable is documented in detail.

Franz Boas, in 1911, noted four distinct roots — for snow on the ground, falling snow, drifting snow, and a snowdrift. He did not suggest this was a large number or that it meant anything in particular. Benjamin Lee

Whorf, in a 1940 article, repeated the observation with seven, three of which he did not source. Later retellings escalated into dozens and then hundreds, each citing the retelling before it. Laura Martin traced this history in 1986, and Geoffrey Pullum's 1989 essay popularizing her case gave it the name it is now known by, the Great Eskimo Vocabulary Hoax. A reader who knows this will discount any argument leaning on the popular version, and should.

Two things are nonetheless true and worth separating out.

Counting words is the wrong question for these languages. They are polysynthetic, building long words from stacked affixes, so the number of constructible "words" for anything is effectively unbounded and a count measures nothing.

And when researchers went and asked — working with speakers who travel on sea ice and need these distinctions to survive — the vocabularies turned out to be both large and highly structured. Surveys across roughly ten Inuit and Yupik dialects have recorded on the order of seventy terms for ice conditions in one Alaskan dialect, around forty snow descriptors in Central Siberian Yupik, and some fifty in a Canadian Inuit dialect. These are not synonyms. They mark whether a surface will bear weight, how it formed, and how it is likely to fail.

So the popular claim is wrong in form and there is something real underneath it. Both halves have to be said, because saying only the first is now the more common error. The example is set aside here not because the underlying observation is false but because the version in circulation cannot be defended, and because better evidence exists.

The sea-ice vocabulary also points at a question that is better supported than the famous one, and it runs in the opposite direction: not whether words shape what speakers notice, but whether what speakers need to notice shapes their words. A tempting extension is that landscapes multiply colour words — more greens where there is forest, more shades of sand in deserts. The evidence points elsewhere. Across the 110 languages of the World Color Survey, warm colours are communicated more precisely than cool ones, and in large collections of photographs the objects people single out are mostly warm-coloured while backgrounds — foliage, sky — are cool. Colour words appear to accumulate where something has to be picked out against its surroundings, and how many a language has tracks how useful colour is to its speakers. The Tsimane' of the Bolivian Amazon, surrounded by green, use colour words less than speakers in industrial societies, and use them more for artificially coloured objects than for natural ones. Where a forest people's colour vocabulary is rich, it can turn out not to be about hue at all. Among the Hanunóo of the Philip-

pines, the term that covers green also means fresh and the term that covers red also means dried out, so a newly cut length of brown bamboo takes the "green" word.

What landscapes reliably multiply is vocabulary for the states of things that matter — the sea-ice terms above, and in deserts the kinds of ground, which geologists found useful enough to borrow: *erg*, *hamada*, *sabkha*, and *wadi* from Arabic, *barchan* from Kazakh, as glaciology took *nunatak*, *pingo*, and *sastrugi* from Greenlandic, Inuvialuktun, and Russian.

Colour has one geographic pattern of its own. Languages spoken where sunlight carries more ultraviolet are more likely to lack a separate word for blue, covering it with their word for green or for dark. One proposed explanation is physical — ultraviolet exposure yellows the lens of the eye and weakens the distinction. An experiment comparing older and younger observers, whose lenses differ in exactly that way, found no difference in how they named blues and greens, which counts against the mechanism; a larger later analysis found ultraviolet exposure to be one influence among several, cultural complexity among them. The pattern is real and its cause is still argued about. What none of it licenses is a claim about any particular language's colour words read off from its landscape — the form of argument this section has just finished taking apart.

What the experiments show

The better evidence comes from studies that test performance rather than count vocabulary.

Colour. Russian has no single word covering the range English calls blue; it obligatorily distinguishes a lighter category and a darker one. In a 2007 study, Russian speakers were faster at telling two blues apart when the pair straddled that boundary than when both fell inside one category. English speakers, for whom the boundary does not exist, showed no such difference.

The elegant part was the control. Participants repeated the task while holding a string of digits in memory, and again while holding a spatial pattern in memory. The Russian advantage vanished under the verbal load and survived the spatial one, which the authors read as the language being recruited during the discrimination itself rather than having permanently restructured perception.

And a 2020 study attempting to replicate it did not find the effect. The later work reported a boundary advantage at a different colour boundary but not at the light-blue/dark-blue boundary the original claim rests on, and found the relevant responses affected by stimulus frequency and presentation order, concluding that the Russian blue categories are less cleanly structured than the earlier result assumed.

That has to be stated here rather than in a footnote, because this is the study the argument of this section was built on. The honest position is now narrower than it was: there is evidence that habitual linguistic categories are recruited during perceptual tasks, the colour case is the most-cited demonstration of it, and the colour case has a serious failed replication. The spatial and numerical evidence below does not depend on it.

Space. Some languages do not use body-relative terms like *left* and *right* for everyday spatial description, using compass-like absolute terms instead — the equivalent of saying there is an insect on one's north arm. Speakers of such languages maintain accurate orientation in conditions where speakers of body-relative languages become lost, and they perform differently on tasks requiring a spatial array to be reproduced after turning around. Habitual use of a reference frame appears to come with habitual tracking of what that frame requires.

Number. Languages with very restricted counting vocabulary look like the sharpest case, and the detail of what was actually found is more interesting than the headline.

The most-discussed example is an Amazonian language two of whose quantity words were initially glossed as *one* and *two*, and later reanalysed by the same researcher as meaning something closer to *small amount* and *larger amount* — with experimental work support-

ing the second reading, since speakers used the supposed word for "one" of as many as six objects.

The quantity experiments found something specific. Speakers matched exact quantities accurately, including large ones, **when the objects were in front of them**. Accuracy collapsed when the array was shown and then hidden before they were asked to reproduce it. The researchers' reading is that the difficulty lies not in perceiving or matching exact quantity but in holding one in memory without a symbol for it.

That is a narrower claim than "they cannot count," and a better one: number words look like a memory technology rather than a perceptual one.

A caution specific to this case. The broader claims made about this language — particularly about grammatical recursion — are vigorously disputed, the researcher's own analysis changed substantially between publications, and very few linguists have field experience with it. The quantity results are the best-supported part of a contested body of work, and the contested parts should not be borrowed to prop up the rest.

What this does and does not establish

The strong version of this idea — that language determines thought, and that what a language lacks words for is unthinkable to its speakers — is not supported and is not claimed here. Speakers readily perceive distinctions their language does not encode, borrow words, coin

them, and describe things by circumlocution. No one is trapped.

The supported version is narrower. Habitually used linguistic categories appear to be recruited during cognitive tasks, and to affect speed and accuracy where the categories are relevant. The effects are modest where they are found.

Replication in this area is genuinely uneven, and the colour case above is not an isolated wobble. The grammatical-gender results have had a mixed record. Effects that look robust in one language pair sometimes fail to appear in another. The pattern is consistent with real effects that are small enough to be fragile, and it is also consistent with a literature that published its positive findings more readily than its negative ones. Both readings remain available, and anyone claiming the question is settled in either direction is ahead of the evidence.

The direction of influence also runs both ways, which complicates any simple story. A community that needs to distinguish ice conditions develops terms for them, and the terms then make the distinctions easier to notice and transmit. Vocabulary is as much a record of what has mattered to a group as a cause of what its members attend to. Asking which came first is usually the wrong question.

Compression, and what it costs

A word is an extreme act of compression. A single short token stands for a category with indefinitely many members, no two identical, and makes that category available for combination with others.

The economy is enormous. It is also lossy in exactly the way chapter 8 described for the genetic code and chapter 13 for perception. Every category discards the differences among the things it covers. Naming a thing makes it easier to think *with* and slightly harder to see freshly, because the name arrives with a set of expectations attached.

This is the same trade made at every level of this book, and language makes it visible because the units are inspectable. The structure of a vocabulary is a record of which distinctions a community found worth the cost of maintaining — and, by omission, which it did not.

Information that outlives its carrier

The consequence that matters most is not individual. Language lets a representation formed in one mind be reconstructed in another, and reconstructed again after the first is gone.

Before that, everything an organism learned died with it, and each generation rediscovered the same things. After it, knowledge accumulates: a practice can be transmitted, refined, and transmitted again, and no

single person needs to hold the reasoning behind it or even to be capable of reconstructing it.

That is the condition for everything in chapter 15, which shows with a specific and consequential example that the mechanism can preserve knowledge nobody involved possesses.

15
Culture, economics, land use, and religion

Culture can be understood as a system of information, practices, beliefs, technologies, and institutions transmitted between people and across generations.

Cultural elements vary, spread, persist, and disappear. Their success depends on usefulness, familiarity, social status, emotional weight, institutional support, environmental conditions, and much else.

This creates parallels with biological evolution, with two differences large enough that the parallel has to be held loosely. Cultural transmission is far faster, and it is not restricted to parent and offspring — a practice can pass between strangers, across populations, and backwards in age. It is also far lower in fidelity. A gene is copied as a sequence; a practice is reconstructed from observation by someone who may have understood it incompletely.

Cultural selection and transmission

Ideas and practices spread through imitation, teaching, storytelling, social pressure, institutions, and communication networks.

Some persist because they solve practical problems. Others persist because they reinforce identity, create cohesion, offer explanation, or become embedded in institutions that have their own reasons to continue.

Cultural systems experience bottlenecks and founder effects. A small group may establish conventions that later become dominant for no reason except that they arrived first. Isolation, migration, conflict, and social boundaries cause traditions to diverge.

The concept of a **meme** is one way of describing a unit of cultural transmission. The word was coined in 1976 by the evolutionary biologist Richard Dawkins, in *The Selfish Gene*, on the model of *gene*, for a tune, a catchphrase or a fashion copied from one mind to another as a gene is copied from parent to offspring. It is useful as shorthand and strained as a mechanism: cultural ideas are frequently not discrete, not copied with anything like genetic fidelity, and not separable from the people holding them.

Knowledge that nobody holds

The most important property of cultural transmission is one the biological analogy obscures entirely: a practice

can carry knowledge that no one performing it possesses.

Maize was domesticated in Mesoamerica and became a staple. It was also, from early on, processed in a particular way before eating: soaked and cooked in an alkaline solution of wood ash or slaked lime, then hulled and ground. The technique is called **nixtamalization**.

How early is worth stating carefully, because the evidence comes in two kinds. Equipment associated with the process has been recovered from Guatemala's southern coast dating to roughly 1500–1200 BCE, which would put it well over three thousand years ago. Direct identification of nixtamalized maize — microscopic residues that can only have come from the process itself — is much more recent, first reported from the Maya site of San Bartolo and dating to the Late and Terminal Classic, around the eighth and ninth centuries CE. The practice is unquestionably ancient; the equipment is suggestive and the direct evidence is late, and the gap between them is a limit on how precisely the claim can be made.

The immediate reasons to do it are obvious to anyone cooking. The hull comes away easily. The grain grinds into a dough that holds together. It tastes and smells better.

The reason that mattered was invisible, and worth getting right, because the standard explanation of it turns out to be wrong.

Maize is poor in niacin and poor in the amino acid tryptophan, which the human body converts into niacin. Without enough of either, a population eating maize as a staple develops niacin deficiency. That deficiency is pellagra: skin lesions, digestive failure, neurological deterioration, and eventually death.

Alkaline treatment prevents it. The usual account of how is that maize contains niacin in a chemically bound form the gut cannot absorb, and that alkali breaks the bond and releases it. That explanation is widely repeated and multiple experiments have failed to support it. The better-supported mechanism is less tidy: the alkali makes maize protein more digestible, which makes its tryptophan absorbable, and the body makes its own niacin from there.

The correction is worth pausing on, given what this section is about. The practice worked for more than a thousand years, and perhaps for over three thousand, without anyone knowing why. Then it was explained, and the explanation was wrong, and the wrong explanation is still the one in general circulation. Knowledge embedded in a practice is hard to recover; knowledge embedded in a plausible story about a practice may be harder still.

Then maize crossed the Atlantic, and this is where the case becomes an experiment that nobody designed. The crop spread rapidly through Europe, Africa, and later the American South, because it was productive and

grew where other grains did not. The processing technique did not travel with it. There was no reason it would have — it looked like a culinary preference, not a nutritional requirement, and nobody on either side of the exchange knew otherwise.

Pellagra followed the crop. Epidemics appeared in northern Italy, in Spain and southern France, in Egypt. In the American South it became a mass health crisis: between 1906 and 1940 it affected more than three million people and killed over a hundred thousand. At its peak in the late 1920s it was among the leading causes of death in the region.

For years it was attributed to an infectious agent, and frequently to the character and habits of the people suffering from it — who were overwhelmingly poor, and whose poverty was the actual reason their diet had narrowed to cornmeal. Dietary cause was established through deliberate investigation in the 1910s and 1920s, against considerable resistance, and the specific vitamin was identified in 1937.

Mesoamerican cooks had solved the problem centuries or millennia before anyone could state what the problem was, and could not have stated it themselves.

That is the thing worth sitting with. The knowledge was real, it was accurate, and it was load-bearing — and it did not exist in anyone's head. It existed in a sequence of actions passed from one person to the next with the instruction that this is how maize is prepared. The

explanation was not merely absent; it was unavailable in principle, since nothing in the available conceptual vocabulary could have contained it.

A practice is a compressed representation of a relationship with the world. It is lossy in a specific and dangerous way: it discards the reason. That makes it cheap to transmit, since a child can learn it by watching and needs to understand nothing. It also makes it fragile, because a practice separated from its reason cannot defend itself. Anyone can see that a step is laborious. Nobody can see what it is for.

This is also a corrective to a familiar condescension. "They did not know why" is often taken to mean "they did not know." The maize case shows these are different claims, and that the second can be false while the first is true.

Economic systems and land use

Economic arrangements develop through the interaction of resources, technologies, institutions, incentives, and behaviour.

Practices suited to particular environmental and material conditions spread. Others become unsustainable and disappear. Agriculture, trade, settlement patterns, property systems, and technologies all shape how land is used and resources organized.

The terms **economic memetics** and **land-use evolution** are used here as ways of examining persistence

and change in these systems. Neither implies that economies or land-use patterns evolve by the mechanisms that operate on organisms. The resemblance is in the shape of the process — variation, differential persistence, transmission — and not in the machinery.

One caution carries over from the previous section. Practices that persist are not thereby good, and they are not thereby understood by those maintaining them. Nixtamalization persisted and was beneficial. Other practices persist because they benefit whoever is positioned to maintain them, because nobody has examined them, or because the cost of changing exceeds the cost of continuing. Persistence indicates that something has been transmitted. It indicates nothing about why.

Religion as an explanatory and social system

Religion can be examined as a cultural system combining explanation, narrative, ritual, symbol, moral framework, and institution. What follows looks at how such systems are organized and transmitted. It does not assess whether any of them is true, and nothing in it should be read as doing so.

Human beings have always faced uncertainty about death, illness, weather, hunger, conflict, disaster, and origins — questions that are urgent, that arrive without warning, and that in most times and places could not be answered by investigation.

Shared narratives do real work under those conditions. They make events interpretable, provide a common frame in which a community can coordinate, carry moral expectation in memorable form, and offer a way of approaching mortality and suffering that can be held collectively rather than alone. Ritual and institution transmit them across generations with far more stability than individual memory allows.

Religion can therefore be examined through several overlapping functions:

- Providing explanations for uncertain or difficult events.
- Establishing shared narratives and identities.
- Supporting cohesion and collective action.
- Transmitting moral expectations and social rules.
- Offering ways to approach mortality, suffering, and purpose.
- Creating institutions and traditions that persist over time.

One experimental finding is worth bringing in here, because it is more specific than the functional list and does not depend on it. Researchers have studied cultural transmission directly by passing material along a chain of people and measuring what survives each retelling — the method chapter 17 returns to. In such chains, content that is *mildly* counterintuitive survives better than content that is either entirely ordinary or wildly

strange. A story in which almost everything behaves normally and one thing does not is retained and retransmitted more faithfully than a story in which nothing is unusual, or in which everything is.

That is a narrow and testable claim about memory and transmission, and it has an obvious bearing on the shape of religious narrative across unrelated traditions — which tends strongly towards the mostly ordinary world containing a few specific exceptions, rather than towards the uniformly strange. It says nothing whatever about whether any such narrative is true. It says something about which narratives, true or false, are likely to still be in circulation.

Two things need saying plainly about the list of functions, because lists of this kind are easy to misread.

The first is that identifying a function is not an explanation of origin, and still less a debunking. The observation that a practice does something useful is compatible with its being true, with its being false, and with the question not being the kind that admits an answer. Many true things are also useful; usefulness is evidence of neither truth nor falsity. The functional analysis here and the question of what is actually the case are different inquiries, and running them together is a mistake regardless of which direction it is run in.

The second is that religious systems are enormously diverse and have no single origin or single function. Treating them as one thing with one explanation would

be the same error the rest of this book warns against — noticing a similarity and assuming a shared mechanism.

Why frameworks talk past each other

One further observation, offered carefully.

Different explanatory frameworks develop their own vocabularies, and vocabularies that grew up answering different questions are rarely interchangeable. Two people may be addressing genuinely overlapping concerns — why there is suffering, what obligations people have to one another, what happens at the end — in terms that do not map onto each other at any point. Mutual unintelligibility is easy to mistake for fundamental disagreement, and conversations frequently fail at the translation step before reaching the substance.

This observation has a limit that matters more than the observation does. It does **not** follow that all disagreements are merely verbal, that every framework is equally correct, or that apparent conflicts dissolve on inspection. Some disagreements are real, some claims are about matters of fact, and some factual claims are wrong. A framework can be internally coherent, socially valuable, deeply meaningful, and mistaken about something checkable, and noting the translation problem does not dissolve any of that.

What the observation is good for is narrow: it suggests that the first question in a disagreement should be whether both parties are using their words the same

way, and that the answer is more often no than it appears.

The interest here is in religion as a form of cultural organization and transmission. The theological and philosophical questions are left open, for the straightforward reason that nothing in this framework is equipped to settle them.

16
The recurring architecture

Sixteen chapters in, the shape of the argument can be set out in one place.

The progression
Physical conditions → chemical networks → compartments → biological systems → metacompartmentalization → nervous systems → language → culture → social and economic systems

This is not a ladder and not a predetermined path. Each level contains its own feedback, constraints, and forms of selection; levels interact in both directions; and nothing required the sequence to continue past any particular point. It is an ordering of what became possible, not of what had to happen.

The same questions at every level

Domain	What varies	What persists	What becomes possible
Physics	Matter, energy, configurations	Stable arrangements; structures maintained by throughput	Dissipative structures — order that exists while driven
Chemistry	Molecules and reaction pathways	Thermodynamically favoured products; self-reinforcing networks	Autocatalysis; collectively self-sustaining sets
Protocells	Compartment contents and membrane composition	Boundaries that localize their own products	Competition between compartments; selection without heredity
Biology	Genetic sequence	Templates copied across generations	Adaptation, specialization, multicellularity, symbiosis
Cognition	Neural states and associations	Memory; compressed representations	Prediction, abstraction, self-modelling
Language	Sounds, words, constructions	Shared convention	Representation transmissible between minds
Culture	Practices, beliefs, technologies	Tradition and institution	Accumulation beyond individual lifetimes

Domain	What varies	What persists	What becomes possible
Economics	Resources, incentives, techniques	Practices that keep working under local conditions	Coordination at scales no participant comprehends
Religion	Narratives, symbols, rituals	Belief and institution	Shared frames for what cannot be investigated

The table compares organizational patterns. It does not claim the rows share a process, and chapter 2's version of this table — which takes a single term and shows the machinery differing underneath it — is the corrective that should be read alongside it.

Where the continuity is mechanical, and where it is not

This is the chapter's real work, because the progression above is the book's central claim and it is not uniformly supported.

Mechanically continuous, from physics to compartments — in principle. Chemistry is physics. Self-sustaining reaction networks are chemistry: chapter 4 shows autocatalysis demonstrated, and small collectively self-sustaining networks built in the laboratory. Whether prebiotic chemistry was rich enough to produce them unaided is, as chapter 4 says, not settled. Compartments form unaided from amphiphiles, and chapter 5's osmotic competition between vesicles is not

an analogy to selection — it is differential persistence under constraint, operating on compartments, observed in a laboratory. Nothing in this stretch requires a new principle at any step; what remains open is whether early Earth supplied the conditions, not whether the chemistry is possible.

A genuine break at heredity. How a chemical system first came to copy a pattern reliably enough for differences to accumulate across generations is not known. Chapters 8 and 17 say so; chapter 12 argues the absence of a single threshold moment is a feature of the category rather than a gap, which is true and does not fill this particular gap. It remains the one place in the progression where the account cannot describe the transition.

Mechanically continuous again, from heredity through cognition. Once templates are copied with variation in a population, standard evolutionary biology carries the account through multicellularity, specialization, nervous systems, and the capacity to build and use internal models. This stretch is the best-supported part of the book and the least original to it.

Weaker at language. That representation became externally transmissible is clear. How, and in what stages, is not, and this book says little about it that would survive a linguist's scrutiny.

Shared vocabulary rather than shared mechanism, above language. Cultural selection, economic persistence, and religious transmission are described here

with the same terms used for chemistry and biology, and the machinery is unrelated. Chapter 2's table makes this explicit for "selection." Chapter 15 makes it explicit for transmission fidelity. The terms are useful for asking the same question of different systems. They are not evidence that the systems answer it the same way.

An honest reader should take the progression as firm in principle in its lower half, broken at one specific joint, firm again through biology, and increasingly a matter of description rather than mechanism towards the top.

What recurs, concretely

Four things appear at nearly every level, and unlike the vocabulary they are structural rather than terminological.

A boundary. Membrane, cell wall, organism, group, institution. Something distinguishes inside from outside, and the distinction is what makes a unit comparable to other units.

A throughput. Nothing on the list maintains itself without energy or material flowing through it. Stop the flow and the organization does not freeze — it disperses. This is as true of an institution as of a convection cell.

Differential persistence. Variants exist and do not last equally. This requires no copying, no competition in any intentional sense, and no population — chapter 5's vesicles demonstrate it with osmotic pressure alone.

Something carried forward. A stable structure, a copied template, a memory, a practice. Without it each cycle starts over and nothing accumulates, which is chapter 11's point.

And a fifth that is easy to miss: **every level has to suppress the one below it.** Chapter 6 made this concrete with cancer — a multicellular organism persists only while it holds down the replication that many of its component cells remain capable of. The same tension recurs with selfish genetic elements inside genomes and with reproductive policing inside colonies. Organization at a higher level is not a state that was reached. It is a cost paid continuously against a lower level that has not forgotten how to act for itself.

Two routes, not one

The progression above describes organization arising by nesting — components incorporated into larger units.

The human microbiome shows a second route. A host and its microbial communities are not nested in that sense; they remain distinct organisms with separate evolutionary interests, exchanging resources and shaping one another's environments. System-level properties arise anyway, from the relationships rather than from incorporation.

Endosymbiosis is what happens when the second route runs to completion and becomes the first. A free-living bacterium associated with a host, then depended

on it, then handed over its genome. The two routes are the same process at different stages, and most relationships never finish the journey.

That matters for reading the progression correctly. Complexity does not only accumulate by things becoming parts of larger things. It also accumulates between things that stay separate — and much of what any organism does is performed by organisms it is not.

17
Important distinctions and limitations

This framework depends on analogy, and analogy must not be confused with mechanism.

A chemical reaction network, a biological ecosystem, a brain, a language, and an economy can all display variation, feedback, persistence, and emergent properties. That does not mean they are governed by identical processes. It may mean no more than that these are the terms available for describing anything that changes.

Several distinctions should remain clear:

- Physical stability is not biological fitness.
- Chemical autocatalysis is not biological reproduction.
- Genetic inheritance is not cultural transmission.
- Neural representation is not symbolic language.
- Cultural persistence is not genetic selection.
- Specialization does not always increase stability.
- Complexity does not necessarily increase over time.
- Emergence does not imply intention or a predetermined direction.

Living systems also do not violate the second law of thermodynamics. They maintain local organization by using energy and exchanging matter with their surroundings, while producing entropy in the larger environment. Local order is compatible with, and in fact paid for by, a larger increase in disorder. Nothing in this book requires an exception to physics, and any argument that appeared to grant one would be an error in the argument.

The strongest objection

An account that only listed its minor caveats would not be worth much. The serious objection to this framework is the following, stated at full strength:

> It is an interesting unifying narrative, and too broad to count as a strong theory. It connects many domains by analogy without isolating sharp, testable mechanisms that would let it be decisively right or wrong. It risks explaining too much with the same pattern. Its key terms — search space, selection, compression — need operational definitions before they can do work. And it generates no predictions that differ from standard accounts in any of the fields it spans.

Each clause is fair, and they are not equally serious.

On analogy. This is partly answered by discipline. Every comparison in this book is supposed to be

labelled as a comparison, and chapters 9, 10, and 14 go out of their way to show where surface similarity hides unrelated mechanisms. Partly answered is not answered. Labelling an analogy does not convert it into an explanation, and a book full of correctly labelled analogies is still a book full of analogies.

On explaining too much. This is the serious one. A pattern that can be found everywhere constrains nothing, and a framework that cannot be embarrassed by any observation is not telling anyone anything. Worse, the capacity to find such patterns is exactly the faculty chapter 13 describes as unreliable and the Preface describes the author exercising at unusual intensity. The person most likely to see variation-selection-retention in every system is the person predisposed to see it, and no amount of care fully corrects for that.

The honest position is that some of this framework's apparent reach is real continuity and some is the vocabulary being elastic enough to wrap around anything. Which parts are which is not fully settled here.

On operational definitions. Fair and largely unmet. "Compression" is used in this book for the genetic code's error tolerance, for perceptual abstraction, for the meaning of a word, and for a cooking technique that carries nutritional knowledge. These have something in common. Whether that something is a measurable quantity or a family resemblance is not established, and the book does not establish it.

On predictions. Fair. Nothing here predicts an observation that standard accounts in chemistry, biology, cognitive science, or anthropology would not also predict. A framework earns its place by forbidding something, and this one currently forbids very little.

Three claims, and what happened when they were checked

The correct response to the last objection is not a defence but a set of claims specific enough to fail. An earlier draft of this chapter offered three, each tied to a particular chapter, and presented them as tests someone might run.

That framing was wrong, and the correction is more interesting than the original. All three have already been investigated, by people who would not describe themselves as working on anything in this book.

1. Environmental cycling increases the retention of prebiotic chemistry. Chapter 3 argues that cycle frequency matters more than elapsed time. This has been tested extensively. Repeated wet–dry cycling drives nucleotides to polymerize into long chains without chemical activation, in conditions resembling those at volcanic hot springs; material that fails to incorporate is washed away while polymers remain, so each cycle both builds and selects. The effect is well enough established that it is now a standard experimental setup

rather than a question. Chapter 3's claim survives, and understates what is known.

2. Compartmentalization improves the persistence of what it encloses. Chapter 5 claims a boundary changes what the chemistry inside is subject to. Also tested. Ribozymes encapsulated in model protocells adapt faster under selection than the same sequences unencapsulated, and compartmentalization protects a cooperative replicating system from being overrun by parasitic sequences that would otherwise destroy it. The vesicle competition described in chapter 5 is a separate result pointing the same way. Claim survives.

3. Symbolic systems show measurable retention biases. Chapters 14 and 15 claim that what survives cultural transmission is shaped by how easily it transmits, not only by whether it is accurate or useful. Transmission-chain experiments — passing material along a line of participants and measuring what degrades — have identified several specific biases. Social content survives better than non-social content. So does content that is mildly counterintuitive. Material tends to lose detail and drift towards what the receiver already expected. Claim survives, and the real findings are sharper than the prediction.

So the framework's three checkable claims were checked, and all three held.

That is a worse result than it appears, and the reason is worth stating plainly. These predictions were not

derived from the framework and then tested. They were assembled by someone who had read around these fields, from directions the evidence was already pointing. A prediction reverse-engineered from results already in circulation is not a prediction. Passing a test chosen after the answer was known demonstrates nothing except that the framework is consistent with things that were already true, which was never in doubt.

What would count is a claim that the framework generates and the fields do not — something this book should predict and a specialist would not expect. The chapter does not have one. Until it does, the objection in the previous section stands undiminished, and the three results above are better understood as the framework being compatible with the evidence than as the framework having earned anything.

Where the account stops

Four places where the framework identifies genuine uncertainty rather than papering over a gap.

The origin of heredity. The largest one, named already at the end of chapter 8. Chapters 3 through 5 describe chemistry that organizes, localizes, and sustains itself, with laboratory demonstrations for each step and nothing yet being inherited. Everything from chapter 8 onward assumes heredity exists. How a chemical system first came to copy a pattern reliably enough for selection to accumulate across generations is not

known, and nothing in this book narrows it. This is not a small remaining detail. It is the transition the whole account is named after and the one it cannot describe.

The hard problem of consciousness. Chapter 13 describes an architecture — recursive self-modelling, distributed subsystems producing an appearance of unity — and that architecture says nothing about why any of it is accompanied by experience. Why there is something it is like to be a system modelling itself is untouched by every word in that chapter. This is not a gap this framework is positioned to close; it may not be the kind of gap that closes by describing mechanisms more thoroughly.

A comment from the author, not a finding: describing this is very self-referential, and difficult. Thinking about writing about consciousness means consciously trying to understand one's own brain in a very specific modality — a brain synthesizing a description of itself while also relying on itself to do so. That last sentence took effort to formulate, and more than ten rewrites before it found the right words; it is worth rereading slowly.

Whether complexity has any direction. The book describes organization accumulating repeatedly. It also insists, in chapter 9, that evolution has no tendency towards complexity. The relationship between those two statements needs more care than it has been given here.

There are two explanations for an apparent increase. One is a systematic drive towards complexity. The other

requires nothing at all: if there is a floor below which organization cannot fall and still persist, then undirected variation alone will produce a distribution that extends further from that floor over time, with no tendency pushing it. Gould's image, from *Full House* (1996), is a random walk beside a wall — the walker drifts away from the wall only because it cannot pass through it, and nothing is pushing.

This is not merely an alternative worth mentioning. Where the question has been examined empirically, the tests have tended to favour the passive explanation and have not found evidence of a drive. The most common organisms on Earth are still single-celled, as they have been throughout, and the conspicuous increase is in the maximum rather than the typical.

A book organized around accumulating complexity has an obvious reason to underweight this, and has probably done so. The honest version is that much of what the later chapters describe as accumulation may be a distribution spreading from a wall, observed by something that happens to be at the far end of it.

Whether collective systems cross the same threshold. Individual minds model themselves. Cultures accumulate models. Whether a population can be said to model its own modelling in any sense beyond the metaphorical is genuinely unclear, and the literature treating it as settled in either direction is unconvincing. *This is the author's speculation and is flagged as such; it is*

included because it follows from the structure of the argument, not because there is evidence for it.

What would count against all of this

A framework should be able to say what would damage it.

Two candidates have to be struck from this list, for the reason given in the section on the three tested claims. *If cycling and compartmentalization made no measurable difference to persistence* and *if cultural transmission showed no systematic retention biases* would both have been fatal — and both have been checked and come out the other way. A falsifier that has already failed to fire is not a falsifier. Listing it is a way of claiming credit for a risk that was never taken.

What remains:

- **If the recurring vocabulary applied equally well to systems generated at random**, it would be describing the describer rather than anything described. Generate structureless systems, hand their descriptions to readers alongside real ones, and see whether the framework distinguishes them. This is the cheapest test available and nobody has run it, including the author.
- **If a single mechanism were found that genuinely did unify these domains**, the book's central hedge — that shared vocabulary is not shared mech-

anism — would be wrong in the opposite direction, and the more confident earlier version of this project would turn out to have been right.

- **If the open gaps closed without the framework contributing anything.** If the origin of heredity is explained by chemistry that owes nothing to search, selection, or compression, then the vocabulary was decoration over the interesting part. This is the most likely of the three.

Three falsifiers, one of them cheap and unrun. That is a thin showing, and it is the accurate one.

The framework is most useful when it identifies a similarity worth investigating and then gets out of the way of the investigation. It is least useful, and most likely to mislead, when the similarity is satisfying enough that nobody checks.

18
Index of terms

Every term this book defines, where it is defined, and its identifier in the companion.

This is an index, not a second glossary: chapter 2 defines the core process vocabulary properly, and *Modern Alchemy*, book 3 of this collection, carries the full apparatus — roughly a hundred and twenty concepts, each with a permanent identifier, a unique symbol, and a one-line gloss. A third partial list between them would be three places to update and one more opportunity for them to disagree.

What was missing, and what this is, is a way to find where in *this* book a term is actually explained.

How to read the third column. Identifiers like `CH-02` are permanent and will not be renumbered; symbols like `Au` are for the companion's notation and are only unambiguous while its glossary is open. Cite the identifier, not the symbol. A dash means the term is used here but is not part of the companion's scheme.

Italic chapter numbers mark a chapter that explains the idea without using the term itself. The term is listed because it is the one a reader is likely to look

up, and the companion uses it. For example, chapter 14 describes sounds combining into words without ever saying *phoneme*.

Core process vocabulary

Defined in chapter 2 and used throughout.

Term	Chapter	Companion
Variation	2	CO-01
Search	2	CO-02
Selection	2	CO-03
Retention	2	CO-04
Amplification	2	CO-05
Concentration	2	CO-14
Compartmentalization	2, 5	CO-15
Metacompartmentalization	2, 6	CO-16
Coupling	2	CO-11
Specialization	2, 6	CO-06
Compression	2, 8, 13, 14	CO-07
Transmission	2, 14, 15	CO-08
Accumulation	2, 11	CO-12
Feedback	2	CO-10
Emergence	2, 6	CO-09
Selection bias	2, 3	CO-13

Physical and chemical

Term	Chapter	Companion
Brownian motion	3	PH-08
Cycles (tides, day–night, wet–dry)	3, 11	PH-10
Thermal and chemical gradients	3	PH-03
Dissipative structure	3, 4, 7	PH-15
Entropy (and why local order is permitted)	17	PH-02
Catalysis	4	CH-01
Autocatalysis	4	CH-02
Autocatalytic set (collective)	4	CH-03
Reaction network	4	CH-04
Formose reaction	4	—
Convection cells	4	—
Belousov–Zhabotinsky reaction	4, 10	CH-06
Amphiphile	5	CH-12
Hydrothermal vent gradients	3, 5	CH-10
Hysteresis	11	—

Protocells and early life

Term	Chapter	Companion
Micelle	5	PL-02
Bilayer vesicle	5	PL-01
Membrane	5	PL-03
Protocell	*5*, 16	PL-04

Term	Chapter	Companion
Electrochemical gradient	5	PH-05
Chemiosmosis (proton pumping)	*5*	PL-11
RNA world	8	PL-05
Ribozyme	8	PL-06
Template replication (copying)	*8*	PL-07
Heredity	8	PL-10
Genetic code; codon; wobble position	8	—

Biological

Term	Chapter	Companion
Natural selection	9	BI-01
Mutation · drift · gene flow	9	BI-02 · BI-03 · BI-04
Adaptation	*9*	BI-05
Convergent evolution	9	BI-14
Vestigial structure	*9*	BI-15
Horizontal gene transfer	9	BI-16
Major transitions	6	—
Endosymbiosis	6	BI-08
Multicellularity	6, 9	BI-09
Distributed labour	6, 7	BI-13
Interdependence	6	CX-05
Microbiome	7	BI-11
Holobiont	7	BI-12
Winogradsky column	7	—
Symbiosis	6, *7*	BI-10

Term	Chapter	Companion
Stratum corneum	7	—
16S ribosomal RNA	7	—
Prion	12	—

Cognition, language, culture

Term	Chapter	Companion
Representation	13	MC-06
Pattern recognition	13	MC-01
Memory · prediction · abstraction	13	MC-02 · MC-03 · MC-05
Strange loop	13	MC-07
Cognitive constraint	*13*, *14*, *15*	MC-08
Language evolution	*14*	MC-09
Phoneme	*14*	MC-10
Articulatory constraint	*14*	MC-11
Symbolic compression	*14*	MC-13
Cultural transmission	*14*, 15	MC-15
Meme	15	MC-16
Cultural bottleneck · founder effect	15	MC-17 · MC-18
Economic memetics	15	MC-20
Land-use evolution	15	MC-21
Myth as explanation	*15*	MC-22
Religious evolution	*15*	MC-23
Nixtamalization	15	—

Cross-domain and speculative

The companion's MF tier, plus the term it builds on, *search space* (CX-10). Everything else here is a framing rather than an established result, and each is marked as such where it appears.

Term	Chapter	Companion
Search space	3	CX-10
Life as search-space explosion	3	MF-01
Life as a cluster of properties	12	MF-04
Sorites structure of the category	*12*	—
Spiral as the geometry of becoming	10	MF-07
Time as repetition plus accumulation	11	MF-08
Language as protochemistry	14	MF-02
Religion as cultural selection	15	MF-03
Culture as cumulative information	14, 15	MF-05
Cross-domain continuity	*1*, 16, 17	MF-06

Terms marked with a dash are used in this book and are not in the companion's glossary — mostly specific experiments, structures, and named phenomena rather than reusable concepts. The companion is a vocabulary for composing ideas, not a subject index, and keeping that boundary is what stops it from becoming a worse version of this chapter.

19
Where these ideas come from

A reading list first, organized by subject, and then the full references.

How the references were checked. Every entry in the reference list at the end of this chapter was checked on 3 November 2026 against the publication record — Crossref, PubMed, or the publisher — for its authors, year, title, venue, volume, pages, and DOI. The check was worth doing: it corrected several details, one of them an error in a publication database rather than in this book.

Whether each work supports the sentence it is attached to is a separate question, and a harder one. For most entries it was checked against the work's abstract, or is settled by the work's own title. The book's most important claims were checked in more depth across four rounds, the last against the works themselves wherever they could be reached. Entries marked † have not had that check yet, usually because the work itself could not be reached or because the relevant detail was found in a secondary account rather than in the work itself.

Four of the checks changed the book's conclusions rather than its details, and each is noted where it occurs: the colour-discrimination replication failure in chapter 14, the status of chapter 17's three claims, the history of the neutron star prediction in chapter 11, and the evidence on the feeling of insight in chapter 13.

A reader who wants to check something should treat a † as a pointer to where to look, not as evidence that the looking has been done. A reader who finds an error here should assume the error is in this chapter rather than in the field.

Energy, order, and structure

The idea that organization can persist as long as energy flows through it, which chapters 3, 4, and 7 all depend on, comes from **Ilya Prigogine**'s work on dissipative structures. His 1977 Nobel lecture states the case directly and is freely available.

Erwin Schrödinger's *What Is Life?* (1944) is the short book that put the thermodynamic question to biologists in the first place, and its guesses about an "aperiodic crystal" carrying hereditary information read remarkably well a decade before the structure of DNA was known.

Alan Turing's 1952 paper on the chemical basis of morphogenesis showed that reaction coupled to diffusion can generate pattern from uniformity — the formal

ancestor of the pattern-forming chemistry in chapters 4 and 10.

D'Arcy Wentworth Thompson, *On Growth and Form* (1917), is the origin of chapter 10's argument that growth rules produce shapes without the shapes being specified anywhere. It is in the public domain.

Douady and Couder's 1992 experiments reproducing phyllotactic spirals in a purely physical system of repelling magnetized droplets carry more of chapter 10 than anything else cited there.

Prebiotic chemistry and the origin of life

Stanley Miller's 1953 report of amino acid synthesis from simple gases and an electrical discharge is the experiment everything in this area is measured against.

The **formose reaction** is older still, traced to Butlerov in 1861. Delidovich and colleagues' 2014 review covers both its autocatalytic mechanism and its selectivity problem, which chapter 4 treats as the more important half.

Stuart Kauffman's *The Origins of Order* (1993) sets out the argument for collectively autocatalytic sets. The subsequent formal work by **Hordijk and Steel** (2004) is more tractable and more recent.

Manfred Eigen's work in the 1970s — a 1971 paper, and the hypercycle papers with Peter Schuster from 1977 — addresses how coupled replicators might sustain one another.

Jack Szostak's laboratory has produced most of what chapter 5 relies on — fatty acid vesicles that grow, divide, encapsulate RNA, and compete osmotically with one another. Much of it is open access.

Günter Wächtershäuser's iron–sulfur world (1988) and the **alkaline hydrothermal vent** proposals associated with **Michael Russell**, **William Martin**, and **Nick Lane** represent the metabolism-first alternative to RNA-first accounts. Lane's *The Vital Question* (2015) is the most accessible entry point.

Lee and colleagues' 1996 report of a self-replicating peptide, from Ghadiri's group, is the demonstration chapter 4 leans on that a fairly simple molecule can catalyse its own formation.

Genetics, the code, and evolution

Francis Crick's wobble hypothesis (1966) and his paper on the origin of the genetic code (1968) frame chapter 8's central question. The second introduced the "frozen accident" position that remains a live competitor.

Freeland and Hurst's 1998 analysis comparing the standard genetic code against randomly generated alternatives is the quantitative result chapter 8 names directly. The follow-up literature, including criticism of the error model used, matters as much as the original.

Charles Darwin, *On the Origin of Species* (1859), remains the clearest statement of the mechanism and is in the public domain.

Lynn Margulis's 1967 paper on the origin of mitosing cells is the modern case for endosymbiosis, and its publication history is part of why chapter 6 tells the story.

John Maynard Smith and Eörs Szathmáry, *The Major Transitions in Evolution* (1995), supplies chapter 6's organizing frame. Their shorter *The Origins of Life* covers the same ground for general readers.

Carl Woese and colleagues' 1990 three-domain paper, and Woese's later work on horizontal transfer and the "Darwinian threshold" (2002), underpin chapter 9's treatment of the base of the tree.

Austin Burt and Robert Trivers, *Genes in Conflict* (2006), catalogues the selfish genetic elements chapter 6 uses to argue that every level of organization must suppress the one below.

Neil Shubin, *Your Inner Fish* (2008), is the accessible account of the path-dependence evidence in chapter 9, such as the jaw bones in the middle ear, and of much the chapter leaves out, including the fish fin in the human hand.

Stephen Jay Gould, *Full House* (1996), makes the argument chapter 17 concedes it has not given enough weight: that apparent increases in complexity can be produced by a random walk with a lower bound and no driving tendency at all.

Spirals in art

Albert Boime (1984) and **Charles Whitney** (1986) read *The Starry Night*'s sky astronomically, independently of each other. The turbulence exchange in chapter 10 is short and unusually instructive: **Ma and colleagues** (2024) make the claim, and **Bourgault and colleagues** (2025) and **Riley and Gad-el-Hak** (2025) take it apart. **Sofaer and colleagues** (1979) describe the Fajada Butte spirals; **Newman and colleagues** (1982) argue the light-casting slabs are a natural rockfall. **Bressloff and colleagues** (2001) give the mathematics of why spirals and lattices recur in visual hallucination, and **Dronfield** (1995) makes the case, contested, that Irish passage-tomb art draws on it.

Microbiology and the microbiome

Grice and Segre's 2011 review of the skin microbiome, and **Byrd, Belkaid, and Segre**'s 2018 successor, carry chapter 7's skin material. Both have freely available author manuscripts.

Naik and colleagues' 2012 work showing that skin commensals tune local immune responses independently of the gut is now the sole support for chapter 7's claim that the microbial community participates in building the barrier.

The 16S ribosomal RNA gene's use as a universal taxonomic marker traces to Woese's work above. Any

current methods review will also cover the limitations chapter 7 lists — copy-number variation, genus-level resolution, amplification bias, and blindness to fungi and viruses.

Mind and language

Douglas Hofstadter, *Gödel, Escher, Bach* (1979) and *I Am a Strange Loop* (2007). The second states the thesis chapter 13 uses more directly; the first is the better book.

Marvin Minsky, *The Society of Mind* (1986), for the complementary argument that the mind was never unified to begin with.

Winawer and colleagues' 2007 study of Russian blue discrimination is the empirical core of chapter 14, including the verbal-interference result that makes the argument work — and **Martinovic, Paramei, and MacInnes**'s 2020 replication attempt, which did not find the effect, has to be read alongside it.

Gibson and colleagues (2017) on why warm colours get more words than cool ones, and **Lindsey and Brown** (2002) with **Hardy and colleagues**' reply (2005) on the geography of words for blue, are the starting points for the question chapter 14 raises in the other direction: what shapes the words, rather than what the words shape.

Stephen Levinson, *Space in Language and Cognition* (2003), for absolute spatial reference frames.

Lera Boroditsky's accessible writing on language and thought, including a widely circulated 2011 piece, is the usual entry point to this literature.

Franz Boas, *Handbook of American Indian Languages* (1911), is where the snow vocabulary observation begins; **Geoffrey Pullum**'s 1989 essay *The Great Eskimo Vocabulary Hoax* documents what happened to it afterwards. Chapter 14 needs both, and also needs the modern documentation of Inuit and Yupik sea-ice terminology that complicates the debunking, collected in *SIKU: Knowing Our Ice* (2010).

Richard Dawkins, *The Selfish Gene* (1976), for the meme concept chapter 15 uses with reservations.

Culture

Robert Boyd and Peter Richerson's *Culture and the Evolutionary Process* (1985) is the standard treatment of transmission and selection in culture.

The **nixtamalization** case in chapter 15 draws on archaeological work on Mesoamerican maize processing, on the nutritional chemistry of alkaline treatment — including the experiments that overturned the bound-niacin explanation still in general circulation — and on the medical history of the pellagra epidemics: three separate literatures. **Joseph Goldberger**'s 1915 dietary trial, which showed pellagra was dietary rather than infectious, is the turning point of the medical story.

Cosmology

All of chapter 11's aside, and all of it contested.

Lee Smolin, *The Life of the Cosmos* (1997), for cosmological natural selection; the original proposal appeared in a 1992 paper. The neutron star mass bound he derived, and the heavier neutron stars subsequently measured, are both documented and both worth reading alongside his response.

Roger Penrose, *Cycles of Time* (2010), for conformal cyclic cosmology. The claimed detection of circular features in the microwave background was published in 2020 and has been disputed; anyone pursuing it should read the rebuttals, not only the claim.

Loop quantum cosmology and the Big Bounce are associated with **Abhay Ashtekar** and collaborators, notably a 2006 paper with Pawlowski and Singh; the technical papers are on arXiv.

These works do not support one unified theory, and several of them contradict each other. They belong to different fields with different standards of evidence and different ideas about what counts as an explanation. That is the condition any framework spanning this much ground has to work in, and pretending otherwise would be the first step towards the failure chapter 17 describes.

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20
Closing perspective

The central idea is that complexity accumulates through interactions among components — particularly when patterns can persist, reinforce themselves, become specialized, and be passed on.

Chemistry finds ways to persist.

Life finds ways to reproduce.

Biological systems find ways to distribute labour and coordinate functions.

Brains find ways to represent.

Language finds ways to transmit.

Culture finds ways to accumulate.

Or, as Ian Malcolm, Jeff Goldblum's chaos theorist in Spielberg's *Jurassic Park* (1993), puts it: "Life, uh, finds a way."

Each of those sentences is doing something slightly dishonest, and it is worth saying so at the end rather than letting the rhythm carry it. None of these systems finds anything. There is no searching and nothing is sought. The phrasing is a compression — the same kind the book has been describing — and like every compres-

sion it discards something, in this case the absence of any agent.

What recurs, and what does not

The honest summary of twenty chapters is that the recurrence is real and uneven.

In some places the continuity is mechanical. Selection among fatty acid vesicles in chapter 5 and selection among early replicators in chapter 8 sit in different rows of chapter 2's table — nothing is copied in the first, a template is copied in the second — but they are physically coupled, in ways that have been measured. A vesicle encloses a sample of the water it forms in and whatever that water holds, RNA included. RNA inside a vesicle makes it grow at the expense of emptier neighbours (chapter 5), and ribozymes inside model protocells adapt faster under selection than the same sequences outside (chapter 17). Nothing is being compared there; each process acts on the other, and the experiments show where they join.

Here the author would go further. Vesicles did not become replicators. They were there all along, forming wherever amphiphiles gathered in water under the right conditions and enclosing whatever was nearby as they closed, or taking it in later, and at some point what was nearby was something that copied: RNA, or something before it that did the same job. On this reading the first cell arose by combination, chapter 6's two evolutionary

histories merging into one: a line of compartments that grew and divided without copying anything, and a line of copies with nothing to hold them together. Cells made the same move again, much later, when they took in the bacteria that became mitochondria and plastids. The membrane gave its catch what open water could not. Chemistry could already draw energy from reactions that pass electrons between molecules; a boundary made another kind possible, a difference between inside and outside, which cannot exist until there is an inside. And it tied the fate of the contents to the wall: a fuller vesicle grows at its neighbours' expense, so whatever kept producing inside was, in effect, building its own boundary. None of this says how copying began. It places the copier outside the vesicle when they met, and where the copier came from remains the open question named at the end of this chapter.

In other places the continuity is a shared description and nothing more. Selection among molecules and selection among ideas are different enough that the word doing duty for both conceals more than it reveals, which is why chapter 2 spends a table pulling them apart.

Most of the book sits between these, and which parts belong where is not fully settled. Chapter 17 says so at length. The difficulty is not that the framework might be wrong in some particular way; it is that a vocabulary this portable is hard to catch being wrong at all, and the

discipline of marking established, contested, and speculative claims is a partial remedy rather than a solution.

One observation about rate

A pattern runs across the whole arc and is worth stating, with its deflation attached.

Each level of organization in this book operates faster than the one beneath it. Geological and cosmic change proceeds over billions of years; biological evolution over millions; cultural change over thousands; technological change over decades.

There is a mechanism for this and it is not mysterious. Each level inherits what the one below it established and does not have to rediscover it. Biological evolution does not re-derive chemistry. Culture does not wait for generations, because transmission between contemporaries is faster than transmission between parent and offspring, and because — as chapter 15 described — a practice can carry knowledge nobody involved possesses. Each layer is building on a substrate that already works, using a faster transmission channel than the layer below had.

Now the deflation. The comparison is less clean than it sounds, because the four categories are not commensurable. "A meaningful change" means something different in cosmology than in technology, and the choice of what counts is made by the person drawing the comparison. A sequence of accelerating intervals can be pro-

duced from almost any history by choosing the milestones accordingly. The mechanism above is real; the tidiness of the series is partly manufactured.

Some accounts take the acceleration further, treating the rate of complexification as significant in itself — as a signature of something rather than a consequence of layered inheritance. That inference depends on the cosmological material in chapter 11, which is fenced off there as speculation for good reason, and it is not drawn here.

What this is for

The purpose of the vocabulary is to make a set of transitions easier to recognize and discuss without assuming they share a mechanism. That is a modest aim, and it is the one the evidence supports.

A reader could finish this book with very different views about what any of it means. Nothing here settles whether the universe has a purpose, whether the patterns point at anything beyond themselves, or what to make of the fact that a configuration of matter on one planet has become capable of asking. Those questions are not answered and are not dismissed. They are a different kind of question, and the honest position is that this framework has no equipment for them.

What it does offer is an account in which the transitions are continuous. Chemistry that persists, boundaries that localize, networks that sustain themselves, pat-

terns that copy, systems that model, minds that tell each other what they have modelled.

One word in that sentence, *continuous*, needs care, because chapters 16 and 17 both say the opposite of what it may sound like. There *is* a break in this account, at heredity, and it is not a small one: how a chemical system first came to copy a pattern reliably is not known, and this book does not narrow it.

The two senses of the word have to be kept apart. A break in the account is a gap in what is understood. A break in the sequence would be a point where something arrived that the preceding step could not supply. The first is real and is named wherever it occurs. The second has not been found anywhere, and every transition that *has* been worked out in detail — amphiphiles closing into vesicles, vesicles competing by osmotic pressure, a model code gaining its tolerance for error from nothing but the order its parts arrived in — turned out to need nothing beyond what was already there.

That is a claim about the transitions that are understood. It is not a promise about the one that is not. An honest account says where its gap is and declines to fill it with either a mechanism or a meaning.

So the book ends where chapter 1 said it would. A reader can finish it holding very different views about what any of it amounts to, and nothing here is designed to prevent that. The questions it leaves open are left open on the page, with the competing answers attrib-

uted and still standing, which is the form it borrowed and the one the material deserved.

It remains an attempt to describe continuity without erasing difference: to notice recurring patterns across the natural and human worlds while staying open to the likelihood that the most important discoveries have not been made yet, and that some of the connections described here will turn out to have been the pattern-finding faculty admiring its own reflection.

The way to tell those apart is to go and check. Chapter 17 lists what checking would look like, including the one test that is cheap and obvious and that nobody — the author included — has yet bothered to run.

What I make of everything

What follows is the author's own rant about what it all means, and the author wants this read as prose rather than a scientific theory.

What I make of everything — much of which was derived from scribblings that made very little sense, and reading accounts of people with schizophrenia or people experiencing manic episodes — point being, take it all with a heavy grain of salt.

There is something greater than "1 human" that is guiding us all. At some point in our past, the human societies that persisted longest essentially collaborated with each other the best, and at some point during evolution

from protocells, we picked up some guiding energies that helped "speed up the evolution process." What these energies are depends very heavily on your culture, surrounding environment, and symbols available to you. So people who experience these occasional moments of "losing touch with reality," they see visions or other experiences that are these energies guiding us, or giving us needed inspiration to help guide humanity and nature towards a more stable and peaceful state. It's like a cell maintaining 'homeostasis,' it's just maintained by humans and a force we don't have the equipment to understand.

Whether that force is God, a holy trinity, a pantheon of gods, nature spirits, planets, or something entirely different — in the end it's all the same. A lot of us are relying on what these forces have been telling us for centuries when we lacked the language to accurately describe them. So in a way, every religion is true — they're just incomplete in the way the human eye throws away ninety-nine per cent of what it takes in. What's in that ninety-nine per cent? Probably whatever is helping to guide us. In a way, saying "the universe is guiding us" might just be my explanation for quite literally everything.

But again, I got these visions in a period of time where I wasn't sleeping and had trouble maintaining focus on finishing sentences. So take it for what you will. Maybe it's the bacteria / holobiont guiding each and every one of us and we're just little carbon-based armour for an immensely complex and orchestrated system of life,

energy, and cooperation. Either way, I won't say which I believe — only a loose theory for my experience on this planet.

About the author

Freddi Van Gogh grew up moving between England, Germany, and the United States, often left to themselves, out in nature or in an empty Oxbridge library, with a great deal of curiosity and a habit of getting lost in books.

Freddi lives a mostly private life, doing science by day and writing fantasy by night. They are not related to "the famous Van Goghs", but have been taken for a lost cousin often enough that they stopped correcting people.

Freddi can be reached at freddi.van.gogh@proton.me, but values their privacy, and so writes under a changed first name.

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